

Stakeholder-Specific Architecture Viewpoints for EU Regulation

A Generative AI-Supported Method

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Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

Abstract

European Union regulation imposes organisational obligations whose implications for individual stakeholders are often difficult to identify. Translating these obligations into representations of daily activities, responsibilities, and dependencies requires a systematic connection between legal requirements and operational evidence. This dissertation proposes a human-supervised method, supported by Generative Artificial Intelligence (GenAI), for identifying, specifying, and materialising stakeholder-specific architecture viewpoints.

Following Design Science Research Methodology (DSRM) and grounded in ISO/IEC/IEEE 42010, the method combines stakeholder interviews, operational-footprint extraction, regulatory relevance analysis, viewpoint specification, and ArchiMate model generation. Traceability links architectural elements and relationships to binding legal text and documented operational evidence, while explicit ownership distinctions, human approval gates, and cross-artifact reconciliation constrain generated content.

The method is demonstrated using the Digital Operational Resilience Act (DORA) through the perspectives of impacted stakeholders. The resulting Architecture Description (AD)s relate regulatory requirements to role-specific work while distinguishing direct responsibilities, external dependencies, and uncertain ownership. Preliminary stakeholder feedback indicates that the formal views can support regulatory understanding and reveal missing operational context, enabling viewpoint refinement. It also identifies limitations in explanatory narratives and simplified diagrams. The demonstrations support the feasibility of the approach, with opportunities for further development. The contribution is a traceable process for communicating stakeholder-specific regulatory impact without assessing organisational compliance.

Keywords

European Union, Conceptual Modelling, DORA Regulation, ArchiMate, Architecture Description, Generative AI

Resumo

A regulamentação da União Europeia impõe obrigações organizacionais cujas implicações para cada parte interessada são frequentemente difíceis de identificar. Traduzir estas obrigações em representações das atividades diárias, responsabilidades e dependências exige ligar sistematicamente os requisitos legais à evidência operacional. Esta dissertação propõe um método supervisionado por humanos, apoiado por inteligência artificial generativa, para identificar, especificar e materializar *viewpoints* de arquitetura específicos de cada parte interessada.

Seguindo a DSRM, com base na norma ISO/IEC/IEEE 42010, o método combina entrevistas às partes interessadas, extração do perfil operacional, análise da relevância regulamentar, especificação de *viewpoints* e geração de modelos ArchiMate. A rastreabilidade associa elementos e relações arquiteturais ao texto jurídico vinculativo e à evidência operacional documentada, enquanto a distinção explícita da atribuição de responsabilidades, as etapas de aprovação humana e a reconciliação entre artefactos limitam o conteúdo gerado.

O método é demonstrado com o DORA, através das perspetivas das partes interessadas afetadas. As descrições arquiteturais resultantes relacionam os requisitos regulamentares com o trabalho específico de cada função, distinguindo responsabilidades diretas, dependências externas e responsabilidades de atribuição incerta. As apreciações preliminares das partes interessadas indicam que as *views* formais podem apoiar a compreensão da regulamentação e revelar lacunas no contexto operacional, permitindo aperfeiçoar os *viewpoints*. Identificam também limitações nas narrativas explicativas e nos diagramas simplificados. As demonstrações sustentam a viabilidade da abordagem, com oportunidades de desenvolvimento. O contributo consiste num processo rastreável para comunicar o impacto regulamentar por parte interessada, sem avaliar a conformidade da organização.

Palavras Chave

União Europeia, Modelação Conceptual, Regulamento DORA, ArchiMate, Descrição Arquitetural, Inteligência Artificial Generativa

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Acronyms

ICT Information and Communication Technology

DORA Digital Operational Resilience Act

eIDAS2 Electronic Identification, Authentication and Trust Services

NIS2 Network and Information Systems

EU European Union

EA Enterprise Architecture

AD Architecture Description

AI Artificial Intelligence

ADL Architecture Description Language

TOGAF The Open Group Architecture Framework

TLPT Threat-Led Penetration Tests

GenAI Generative Artificial Intelligence

ID Identifier

CSIRT Computer Security Incident Response Team

LLM Large Language Model

NIST National Institute of Standards and Technology

RE Requirements Engineering

UML Unified Modelling Language

RAG Retrieval-Augmented Generation

NLP Natural Language Processing

MURCIA MUlti-layered Regulatory Change Identification and Analysis

EoI Entity of Interest

CSA European Union Cybersecurity Act

XML Extensible Markup Language

AAC Architecture-as-Code

DSL Domain Specific Language

HTML HyperText Markup Language

DSRM Design Science Research Methodology

TFEU Treaty on the Functioning of the European Union

MAD Multi-agent debate

RTS Regulatory Technical Standards

ITS Implementing Technical Standards

AKS Azure Kubernetes Service

1

Introduction

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1.1	Context and Motivation	2
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The European Union (EU) had a surge in cybersecurity legislation in recent years, creating many granular regulations, one of which the Digital Operational Resilience Act (DORA). This fragmented landscape creates a significant architectural challenge for organizational governance, where compliance serves as a primary constraint shaping organizational structures and technical dependencies.

A critical gap remains between descriptive legal prose and the formal models required for operational management. Traditional manual Requirements Engineering (RE) is increasingly inadequate for managing the volume and complexity of these concurrent obligations. This work addresses this gap by proposing a refined, viewpoint-driven architectural method grounded in ISO/IEC/IEEE 42010 and accelerated with the use of Generative Artificial Intelligence (GenAI). The following subchapters introduce the project context and motivation, objectives, contributions, and the structure of this dissertation.

1.1 Context and Motivation

The legal framework is a major constraint for all enterprises, notably when dealing with an increasing number of legal regulations or when their complexity rises, as in the financial sector [4].

Recent legislative initiatives, such as DORA, impose extensive compliance mandates on organizational processes, governance structures, and information technology landscapes. While these regulations are formulated as legal texts with compliance requirements, their impact is fundamentally operational.

Translating abstract regulatory prose into actionable organizational change presents a critical enterprise architecture challenge. Traditional compliance approaches typically generate monolithic checklists or isolated requirement spreadsheets. While these artifacts capture legal obligations, they fail to address the diverse concerns of individual organizational stakeholders, such as Chief Information security officers, enterprise architects, or even developers. Each stakeholder requires visibility into how a given regulation directly impacts their specific operational footprint, which organizational assets or capabilities are affected, and the exact nature of that impact.

Enterprise Architecture (EA), and specifically the ArchiMate modelling language, offers a rigorous foundation for modelling complex systems and stakeholder concerns through the concept of viewpoints. Applying this foundation to EU regulatory contexts creates a methodological challenge: translating legal requirements and stakeholder-specific operational concerns into justified viewpoints and tailored architectural views.

This dissertation is the spiritual successor of the Master's thesis *Conceptual Modelling of Legal Compliance Requirements – The cases of Electronic Identification, Authentication and Trust Services (eIDAS2) and Network and Information Systems (NIS2)*. That work established a methodical approach for structuring legal requirements into formal Architecture Description (AD)s using the ArchiMate modelling language. Its baseline method combined an Excel-based traceability matrix with a hierarchical numbering system to link legal clauses to architectural elements. It also explored the use of generative services for review and documentation, including unit-based reviews with ChatGPT 5 and AI-enhanced JArchi integration using models such as LLaMA and Gemini.

Building on these foundations, this dissertation proposes a method for identifying, defining, and instantiating stakeholder-centric viewpoints in the context of EU regulation, with DORA as its regulatory scope. GenAI is used as an accelerator and method stressor to support the controlled creation and review of ADs. By formalising stakeholder concerns in relation to evidenced operational context and relevant legal requirements, the method guides the generation of ArchiMate views that represent the regulation's stakeholder-specific footprint. The resulting artefacts are intended to provide clear, role-tailored transparency into the responsibilities, dependencies, and regulatory impacts relevant to a stakeholder's operational domain.

It is fundamental to stress that this work does not aim to provide authoritative legal interpretations of EU legislation, automate compliance decisions or certification processes, replace legal or architectural expertise with artificial intelligence, or define new regulatory requirements or assess organizational compliance. GenAI is used exclusively as a support tool and a method stressor for identifying ambiguities, not for decision making.

1.2 Objectives

The primary objective of this work is to develop a method that identifies and formalizes stakeholder-specific viewpoints, aligned with the ISO/IEC/IEEE 42010 standard, to systematically translate legislative requirements into tangible operational footprints using the ArchiMate modelling language.

In accordance with Design Science Research Methodology (DSRM), this research pursues this objective by designing and demonstrating a purposeful artifact: a human-supervised methodology for discovering, specifying, and materialising stakeholder-specific, regulation aware ArchiMate viewpoints [5]. The artifact comprises a repeatable process, explicit evidence and traceability rules, validation gates, and a GenAI-supported implementation. The resulting architectural descriptions are instantiations of the artifact, rather than the sole research contribution.

The secondary objective is to create ADs that are sufficiently comprehensible for their intended stakeholders to inspect, challenge, and enrich the viewpoint from which they are derived. Stakeholder feedback may introduce corrections, additional operational context, concerns, or scope refinements, which can be incorporated into the viewpoint. In this way, a generated architecture description becomes not only a communication artefact, but also a feedback mechanism for improving the quality and relevance of subsequent viewpoint generations.

The research proposition of this thesis is that a human-supervised methodology which anchors regulatory requirements in binding legal sources, connects them to documented stakeholder operational evidence, and applies explicit scope, relationship, and approval rules can systematically derive stakeholder-specific ArchiMate viewpoints. The resulting architectural descriptions preserve the provenance of represented requirements, operational facts, and architectural relationships, while making relevant responsibilities, dependencies, and regulatory context of the stakeholder available for structured inspection.

Furthermore, the proposition holds that such descriptions can function as feedback artefacts: an intended stakeholder can use them to identify omissions, corrections, additional concerns, or relevant operational context, which may be incorporated into later iterations of the viewpoint.

1.3 Contributions

Guided by DSRM, this research identifies a regulatory-modelling problem, defines objectives for a purposeful artefact, designs and develops that artefact, and demonstrates its application in the DORA context [5]. The contributions of this dissertation are therefore methodological and artefactual. They do not constitute an authoritative legal interpretation of DORA, an automated compliance assessment, or evidence of organisational compliance.

1. **An evidence-grounded methodology for stakeholder-specific regulatory viewpoints.** The thesis contributes a repeatable method for discovering stakeholder operational context, identifying regulatory relevance, defining viewpoint scope, and materialising stakeholder-specific ArchiMate viewpoints and views. The method connects regulation-aware modelling to the documented responsibilities, dependencies, concerns, and operational environment of the stakeholder.
2. **A dual-source traceability and evidence-control approach.** The methodology establishes explicit links from architectural elements and relationships to both binding legal sources and documented operational evidence. It introduces rules for scope, relationship selection, ownership, and approval that prevent unsupported legal or architectural inferences from being silently incorporated into the resulting artefacts.
3. **A feedback-oriented architecture-description process.** The generated ADs are designed not only to communicate regulatory impacts to their intended stakeholders, but also to support structured feedback. Stakeholders may inspect a view and identify missing context, incorrect assumptions, additional concerns, or scope refinements, which can subsequently enrich the viewpoint specification and inform later generations.
4. **A DORA-based demonstration of the methodology.** The thesis applies the method to representative DORA stakeholder scenarios, producing traceable ArchiMate artefacts and accompanying explanatory material. This demonstration shows how the methodology can be instantiated in a financial-sector regulatory context, while deliberately not claiming formal external validation, legal correctness, or unrestricted generalisability.

1.4 Document Structure

The remainder of this dissertation is organised as follows. The dissertation first establishes the conceptual and legal foundations required by the proposed method. Chapter 2 introduces conceptual modelling and enterprise architecture, including architecture descriptions, viewpoints and views, ISO/IEC/IEEE 42010, ArchiMate, and the role of generative language models in modelling work. Chapter 3 then

presents the relevant legal context, explaining the structure of EU normative acts and introducing DORA as the regulation used in the demonstration.

The research is situated in relation to existing work in Chapter 4, which reviews regulatory modelling, automation, Large Language Model (LLM) agents and skills, multi-agent debate, and the previous thesis from which this work develops. The problem analysis in Chapter 5 uses these foundations to identify the methodological need for stakeholder-specific, evidence-grounded regulatory viewpoints and controlled GenAI support.

The resulting methodology is presented in Chapter 6. It discusses the evolution of the method and describes the complete pipeline through which stakeholder context, legal requirements, and architectural artefacts are produced and reconciled. Chapter 7 demonstrates the method in the DORA context through architecture descriptions from representative banking-stakeholder perspectives. Finally, Chapter 8 concludes the dissertation by summarising its outcomes, discussing limitations, and outlining future work.

2

Conceptual Modelling

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This chapter presents the conceptual foundations that support the methodology proposed in this dissertation. It first relates conceptual modelling to EA, before introducing architecture descriptions, viewpoints, and views through the concepts established by ISO/IEC/IEEE 42010. The chapter then presents ArchiMate as the modelling language used to express the resulting architectural representations and discusses the opportunities and limitations of using generative language models in modelling work. Together, these foundations establish the terminology and modelling principles used throughout the remainder of the dissertation.

2.1 Conceptual Modelling and Enterprise Architecture

Conceptual modelling is about describing the semantics of software applications at a high level of abstraction. Specifically, conceptual modellers:

- Describe structure models in terms of entities, relationships, and constraints;
- Describe behaviour or functional models in terms of states, transitions among states, and actions performed in states and transitions;
- Describe interactions and user interfaces in terms of messages sent and received and information exchanged.

Although these activities are often associated with individual software applications, they are also applicable to the wider enterprise domain. Enterprise conceptual models support the understanding and communication of organisational situations by representing relevant business, organisational, information, and technical concepts at an appropriate level of abstraction. Anaby-Tavor et al. report that, in their empirical study of practice at IBM, business analysts, business architects, and solution consultants used modelling-related practices and work products to understand business contexts. Such work products could subsequently inform formal software-requirement definitions or recommendations for business activities [6].

At this broader scale, conceptual modelling and EA are complementary. Conceptual modelling provides the means to identify and express the relevant concepts, relationships, and constraints of an enterprise. EA provides the enterprise-wide purpose for these representations: it concerns the integrated relationships among organisational structures, business processes, information systems, and technical infrastructure. Rather than treating organisational structure, business processes, information systems, and infrastructure as independent domains, EA provides an integrated perspective through which their relationships can be considered together. [7].

Consequently, an enterprise model should not attempt to reproduce every detail of an organisation. Its value lies in selecting the concepts and relationships that are relevant to a given analytical or communicative purpose, while preserving sufficient context for those selections to be interpreted. This principle is particularly important where a change, obligation, or risk crosses organisational and technical boundaries: its significance may depend on the relationship between responsibilities, processes, information, applications, and infrastructure rather than on any one element in isolation.

ISO/IEC/IEEE 42010 recognises enterprises as entities whose architectures may be expressed through structured architecture descriptions [1]. It therefore provides an appropriate conceptual foundation for using enterprise models to address architectural concerns without confusing a representation with the enterprise itself. In this dissertation, ArchiMate supplies the EA modelling language used to ex-

press selected concepts and relationships in a consistent notation [8]. The following sections introduce the AD concepts and ArchiMate foundations that guide this use.

2.2 Architectural Descriptions, Viewpoints and Views

In their typical usage, conceptual-model diagrams are high-level abstractions that enable clients and analysts to understand one another, enable analysts to communicate successfully with application programmers, and in some cases automatically generate (parts of) the software application¹. These diagrams are implementation-independent and represent a domain's key concepts and their relationships. By abstracting away technical details, conceptual models capture the universe of discourse for a system or organization.

Conceptual modelling ensures clarity in every system design, being essential for creating clear ADs, which formally represent system architectures [3]. As written in ISO 42010, an AD is an "work product used to express an architecture", and not the architecture itself. An AD may "take the form of a document, hypertext or a wiki, a collection of models, a repository, or some other medium"² and addresses stakeholder concerns through views and viewpoints [3]. Additionally, an AD is made up of elements, which can be "stakeholders, concerns, stakeholder perspectives, aspects, other ADs (correspondences), architecture views, view components, architecture viewpoints, and model kinds" [1].

These concepts are formalised in ISO 42010. The viewpoint is a set of rules used to create a view: "A set of conventions for constructing, interpreting, using and analysing one type of Architecture View. A viewpoint includes Model Kinds, viewpoint languages and notations, modelling methods and analytic techniques to frame a specific set of Concerns." These viewpoints may express operational, technical, logical or other types of concerns². According to The Open Group Architecture Framework (TOGAF), a viewpoint can also be seen as a schema for a kind of architecture view. It establishes the conventions for constructing an using a view to address a specific concern (or set of concerns) about a system-of-interest [9]. The ArchiMate 3.2 Specification also provides a definition of viewpoints. They "define abstractions on the set of models representing the EA, each aimed at a particular type of stakeholder and addressing a particular set of concerns". Viewpoints can also be used to view certain aspects in isolation, or to relate two or more aspects [8]. For example, a governance viewpoint can concentrate on legal accountability and decision rights while an operational viewpoint can show processes, actors, information, and service dependencies. These are suitable choices for this work. A single viewpoint can govern one or more views, allowing the same modelling conventions to be reused at different levels of detail or for different stakeholder audiences [1].

A view is an "information part comprising portion of an architecture description" [1] that "expresses

¹Taken from ConceptualModeling.org, available at <https://conceptualmodeling.org/ConceptualModeling.html>

²Taken from iso-architecture.org, available at <http://www.iso-architecture.org/ieee-1471/index.html>

the Architecture of the Entity of Interest (EoI) from the perspective of one or more Stakeholders to address specific Concerns, using the conventions established by its viewpoint"². The EoI refers to the subject of the AD, which can range from software and systems to entire enterprises.

Deriving from these definitions, it can then be said that a viewpoint is a set of rules used to create, interpret, analyse and use one type that addresses a specific set of concerns (representing different stakeholder) about an entity of interest. These concerns may take different forms, such as operational, technical or logical.

An Architecture Description Language (ADL) is a language that enables enterprise architects to create comprehensive representations of its ADs. An example of this is ArchiMate, a "comprehensive enterprise data modelling language that allows architects to create clear and integrated models of enterprise architecture"³. The ArchiMate specification emphasizes modelling with concepts, offering a comprehensive way to represent abstract structures and their relationships [3]. As an ADL, ArchiMate "provides a set of entities and relationships with their corresponding iconography for the representation of Architecture Descriptions" [8].

2.3 ISO/IEC/IEEE 42010

ISO/IEC/IEEE 42010:2022 provides an international basis for the structured expression of architectures. Its purpose is not to prescribe how an organisation must design a system, select a technology, or interpret a legal obligation. Instead, it specifies requirements for the structure and expression of ADs for a broad range of entities, including software, systems, enterprises, systems of systems, products, services, technologies, and business domains [1]. ISO/IEC/IEEE 42010 offers a way to organise an architecture description of those relationships without confusing the description with the legal acts or with the architecture itself.

The distinction between an architecture and its description is fundamental. An architecture concerns the essential characteristics of an entity in its environment, whereas an AD is the work product used to express that architecture [1]. In the context of legal compliance, an AD is therefore an analytical and communicative artefact. It can show how a regulated organisation, a national implementation environment, or a cross-border service ecosystem is arranged in relation to legal obligations. It does not replace the binding text of the law, determine the final legal meaning of a provision, or prove that an organisation is compliant. Its contribution is to make the assumptions, responsibilities, dependencies, and evidence relevant to compliance visible to the people who need to inspect and discuss them.

For this purpose, the entity of interest must be defined with care. It may be an organisation subject to DORA or another bounded implementation context. Once the entity is identified, the architecture

³Taken from leanix.net, available at <https://www.leanix.net/en/wiki/ea/enterprise-architecture-frameworks>

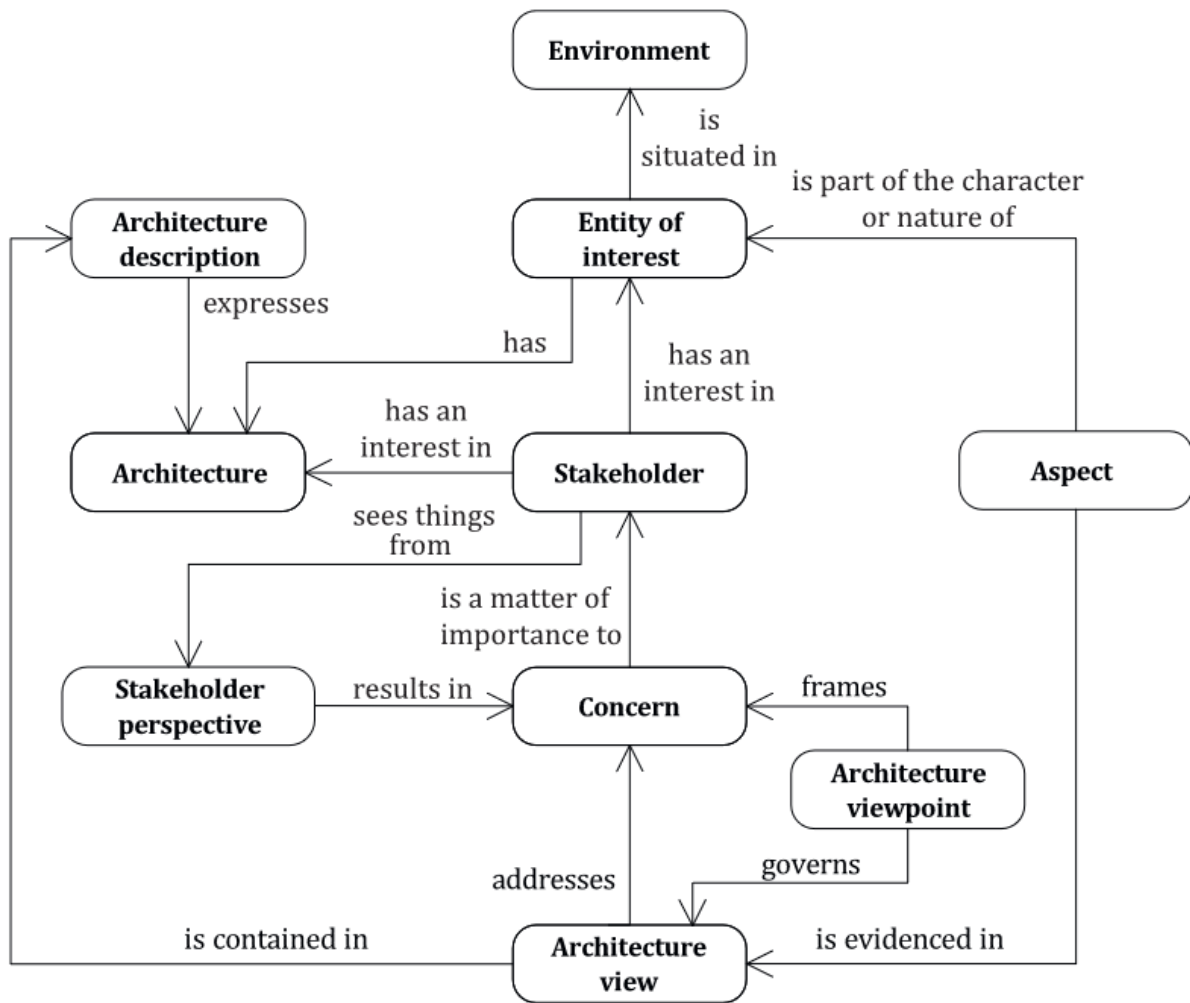


Figure 2.1: Relationships among the central ISO/IEC/IEEE 42010 architecture-description concepts [1].

description can identify its stakeholders and their concerns. Stakeholders may include competent authorities, regulators, enterprise architects, security managers, service providers, relying parties, system operators, auditors, and citizens. Their concerns are not identical: an authority may need assurance that responsibilities and reporting paths are clear, while a service provider may need to understand data exchanges, controls, and dependencies. ISO/IEC/IEEE 42010 treats concerns as matters of relevance or importance to stakeholders and explicitly accommodates legal and regulatory considerations among the matters that can affect an architecture [1]. This makes stakeholder and concern identification a substantive modelling activity rather than a preliminary administrative step. The relationships among these central ISO/IEC/IEEE 42010 concepts are summarised in Figure 2.1.

ISO/IEC/IEEE 42010 also supports the disciplined use of AD frameworks, AD languages, model kinds, perspectives, aspects, and correspondences. A framework can capture conventions shared by a community or domain, while an AD language supplies the means through which models are ex-

pressed [1]. In the present context, a framework can make explicit how legal provisions are selected, how obligations and constraints are represented, how stakeholder concerns are grouped, and how models are checked for consistency. Correspondences are particularly valuable because they record named relationships among elements of an AD [1]. A correspondence record can relate a legal provision to the architectural elements that interpret or operationalise it, the views in which those elements appear and the evidence supporting the interpretation. In this way, traceability becomes more than a visual connection in a diagram: it becomes a maintained explanation of why an element is present and how it relates to the authoritative source.

ArchiMate can provide the modelling language through which selected viewpoints are materialised. The language is intended to describe, analyse, and visualise relationships among business processes, organisational structures, information flows, applications, and technical infrastructure [8]. It is consequently well suited to show the operational consequences of requirements derived from DORA, including the relationships between actors, services, processes, information objects, and technology. Nevertheless, ArchiMate is a modelling choice rather than a notation mandated by ISO/IEC/IEEE 42010. The latter deliberately does not prescribe methods, model types, notations, techniques, tools, formats, or media for creating and maintaining an AD [1]. ArchiMate diagrams, traceability matrices, legal-source registers, and explanatory prose should therefore be treated as complementary parts of one architecture description, each selected because it is useful for the concern and stakeholder being addressed.

This distinction also clarifies the relationship between ISO/IEC/IEEE 42010 and ISO/IEC/IEEE 42020. ISO/IEC/IEEE 42010 concerns the structure and expression of architecture descriptions. ISO/IEC/IEEE 42020 establishes process descriptions for the governance and management of collections of architectures and for the architecting of entities [1, 10]. The standards may be used together: ISO/IEC/IEEE 42020 can guide how an organisation conducts activities such as conceptualisation, evaluation, elaboration, and governance, while ISO/IEC/IEEE 42010 can guide how the resulting AD communicates its architecture. Maintaining that separation is important in legal-compliance modelling. It prevents a description of legal and operational relationships from being mistaken for a complete architecting method, and it preserves the intended role of the AD as a stakeholder-centred, traceable, and reviewable representation of the architecture relevant to regulatory compliance.

2.4 ArchiMate

ArchiMate is a standard enterprise-architecture modelling language developed by The Open Group. It provides a common vocabulary and notation for describing, analysing, and communicating the structure and operation of an enterprise across business, application, and technology domains. Its purpose is not to describe implementation detail exhaustively, but to provide a coherent architectural overview that can

be understood by different stakeholders⁴.

An ArchiMate model is an abstraction of an enterprise rather than the enterprise itself. Views select the concepts and relationships relevant to particular stakeholder concerns, while retaining a consistent underlying metamodel. This makes it possible to relate organisational responsibilities, processes, information, software, infrastructure, goals, and change initiatives within one modelling language.⁴

2.4.1 Elements and Relationships

ArchiMate distinguishes three fundamental aspects of an enterprise: active structure, behaviour, and passive structure. Active-structure elements are entities capable of performing work, such as a Business Actor, Application Component, or Node. Behaviour elements describe what these entities do, for example a Business Process, Application Function, or Technology Service. Passive-structure elements represent information or other objects that are used, created, or transformed, such as a Business Object, Data Object, or Artifact. These aspects recur across the Business, Application, and Technology layers, allowing equivalent architectural questions to be expressed at different levels of abstraction⁵. [8]

Relationships give the model its meaning by specifying how elements are connected. Structural relationships describe responsibility, implementation, and whole–part structures: Assignment allocates responsibility or behaviour; Realization links a concrete element to a more abstract one; Composition and Aggregation express different forms of grouping. Dependency relationships include Serving, where one element provides functionality to another, and Access, where active structure or behaviour reads or writes passive structure. Dynamic relationships describe change and exchange: Triggering represents temporal or causal precedence, while Flow represents the transfer of information, material, or value. Influence links motivation concepts, and Association is reserved for relationships that cannot be expressed more precisely⁵. [8]

The choice of relationship must therefore reflect the meaning evidenced in the architecture. The metamodel constrains which relationship types are valid between particular elements, thereby preserving semantic consistency.⁴

2.4.2 Colours and Notation

A crucial distinction is that colour is not part of ArchiMate semantics. The ArchiMate guidance explicitly leaves colour selection to the modeller. Therefore, colour must not be used to redefine an element's type or to imply a relationship that is absent from the model. Its role is communicative: it can make layer boundaries, scope, changes, or selected analytical concerns easier to perceive.⁴

⁴Taken from ArchiMate 101: A Practical Introduction, from the ArchiMate community, available at <https://archimate-community.pages.opengroup.org/workgroups/archimate-101/>

⁵Elements and relationships can be viewed in the ArchiMate reference cards <https://anyflip.com/nnfi/rdht>

Table 2.1: Pipeline colour convention by ArchiMate layer

Layer	Pipeline colour	Interpretation
Business		Light yellow
Application		Light blue
Technology		Light green
Motivation		Light purple

The methodology presenting in chapter 6 of this work follows the palette presented in 2.1. This convention aligns with the widespread visual association of business with yellow, application with blue, and infrastructure/technology with green, but it is a project convention rather than a mandatory ArchiMate rule. This is synthesized in table 2.1. The European Interoperability Reference Architecture similarly uses these layer associations while declining to impose colour rules. [11]

Colour should be redundant with notation. Element icons, labels, relationship line styles, and a visible legend must still distinguish concepts in monochrome printing or for colour-vision deficiencies.

2.5 Generative Language Models in Modelling Work

In the context of EA and regulatory modelling, generative LLMs function primarily as engines for abstraction, comparison, requirements elicitation and validation, and identifying “semantic deltas” [12, 13]. These models provide a mechanism to navigate the “abstraction gap” between unstructured legal prose and the formal constraints required by architectural descriptions. By performing semantic abstraction, generative LLMs can synthesize complex regulatory provisions into high-level summaries or translate them into the structured syntax required by ADLs [14]. Further more, they facilitate multi-layered comparison by identifying “semantic deltas”—changes in legal meaning that go beyond simple textual differences [12]

However, the integration of GenAI into formal architectural practises is constrained by three critical limitations:

- **Hallucination:** Despite their fluency, models are prone to generating ungrounded assertions that appear plausible but have no factual basis in the source regulation [15];
- **Opacity:** As defined in ISO 42010, an architecture rationale “records explanation, justification or reasoning about architecture decisions”, meaning that logic behind decisions must be explicit and recorded in contrast with GenAI, that often functions as a “black box”, not showing the logic behind the generation.
- **Traceability:** The non-deterministic nature of GenAI complicates the ability to maintain a persistent, verifiable link between a generated architectural component and the specific paragraph of a

regulation [13].

Given these constraints, generative LLMs are positioned in this work as method support and stressors, rather than autonomous analysts. They serve as assistants that augment the human architect's ability to manage regulatory volume and complexity. Additionally, they act as “stressors” to traditional modelling methodologies. Their introduction forces a re-evaluation of how models are validated and how trust is established in an architecture description.

3

European Normative Acts

Contents

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Every action taken by the EU is based on treaties. They are the starting point for EU law and are hence referred to as primary law. The EU is limited by the principle of conferral, meaning that the EU can “only act within the limits of the competences conferred upon it by the Member States in the Treaties to attain the objectives set out therein. Competences not conferred upon the Union in the Treaties remain with the Member States.” [16]

The body of law that comes from the principles and objectives of the treaties is known as secondary law. These include legislative and non-legislative acts. Currently there are five types of juristic acts, including regulations, directives, decisions, recommendations and opinions¹.

- Regulations - Legal acts that are binding in their entirety. They are directly and uniformly applicable to all EU member states as soon as they enter into force, without needing to be transposed into

¹Types of EU law can be found here: https://commission.europa.eu/law/law-making-process/types-eu-law_en

national law.

- Directives - Set binding objectives upon EU member states to achieve a certain result, but leave them free to choose how to achieve these objectives.
- Decisions - Binding in their entirety, can specifies those to whom it is addressed, being binding only on them.
- Recommendations - Non-binding. They allow the EU institutions to make their views known and to suggest a line of action without imposing any legal obligation.
- Opinions - Also non-binding, opinions allow EU institutions to make a statement without imposing any legal obligation.

As a practical example, eIDAS2, European Union Cybersecurity Act (CSA) and DORA are regulations, hence having a legal binding that applies uniformly across all EU member states. NIS2 on the other hand, as a directive, sets goals that member states must achieve in the way they see fit, allowing the country to implement measures that lead to these objectives through national law.

3.1 Regulation as an Architectural Constraint

Regulation functions as a major architectural driver. As noted by Bolinhas [3], “the legal framework is a significant constraint for all businesses, notably when dealing with an increasing number of regulations and directives” [4]. These constraints may affect different dimensions. Regulations like DORA mandate the creation of specific organizational units, or roles, such as the Lead Overseer. This effectively “draws” the organizational map, implementing a structural constraint. These constraints also exist in a behavioural form, since legislations describe rules, which are fundamentally rights and obligations [17] describing what can and cannot be done. Other constraints, like data protection laws, impose constraints on how to handle business objects, defining how data can be stored, for how long and who has access rights to the data. DORA also imposes a type of “supply chain” dependencies, requiring financial organizations to audit and constrain their third-party service providers [18].

Organisations may be subject to several interacting legal requirements, whose implementation can constrain architectural choices. ADs can make these constraints and dependencies explicit and support comparison between implementation alternatives within the applicable legal boundaries. Resolving conflicts between regulations is outside the scope of the demonstrations presented here, which examine stakeholder-specific representations of DORA.

DIRECTIVE (EU) 2022/2555 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL

of 14 December 2022

on measures for a high common level of cybersecurity across the Union, amending Regulation (EU) No 910/2014 and Directive (EU) 2018/1972, and repealing Directive (EU) 2016/1148 (NIS 2 Directive)

(Text with EEA relevance)

Figure 3.1: Title of the NIS2 directive

3.2 Legislation Structure

Identifying the structure of an EU legal act is necessary for locating the provisions represented by architectural requirements. The Interinstitutional Style Guide describes the principal components of an act²: [19]

1. **Title;**
2. **Preamble**(Citations, Recitals, Solemn Forms);
3. **Enacting Terms** (and Annexes).

Presenting each structural part in order, and starting with the title, that already holds a lot of important information:

- The type of act (referenced in 3);
- Institution(s) that adopted the act (Parliament, Council, Commission);
- Whether the act is amending any existing law, and lists all the acts that are amended or repealed;
- If it is an implementing or delegated act;
- The date of adoption;
- The reference number of the act;
- Indication of the broad policy area (EU, Common Foreign and Security Policy, or European Atomic Energy Community);
- Outlines the subject/contents of the act, sometimes also providing a short title added at the end.

Figure 3.1 presents the title of the directive NIS2, which unlike DORA provides all of the informations listed above.

²Structure and a guide on how to read EU law can be found on: <https://fabianbohlenberger.com/2024/04/10/how-to-read-eu-law/comment-page-1/>

Next to the title in the legal text is where the preamble is found. It encapsulates everything between the title and the “Enacting terms”. The preamble is made up of three main elements: citations, recitals and solemn forms.

- **Citations** establish the legal authority of the EU and detail the legislative procedure used for adoption, relying on standardized language that generally starts with “Having regard to...”. DORA establishes the necessary legal authority by citing Article 114 of Treaty on the Functioning of the European Union (TFEU) in the preamble (“Having regard to the Treaty on the Functioning of the European Union, and in particular Article 114 thereof”) [18];
- **Recitals** then supply the statement of reasons, specifying the measure’s goals, the need for EU action, and the reasoning behind the chosen regulatory method. These are numbered and either begin with the word “whereas”, or are placed after a single “whereas” before the enumeration begins. They provide interpretative value but are non-binding.
- **Solemn forms** act as grammatical brackets, turning the entire preamble into a single continuous sentence. It opens by identifying the enacting institutions (e.g., “THE EUROPEAN PARLIAMENT AND THE COUNCIL OF THE EUROPEAN UNION,”), encompasses the citations and recitals, and concludes with the enacting formula: “HAVE ADOPTED THIS REGULATION:”.

Lastly, the enacting terms are the part of the act that sets the new rules. Enacting terms are composed of numbered articles that can be grouped into sections, chapters, titles and parts (in this order, listed from the greatest granularity level to the least). Any of these groupings can contain articles, which may also be supplemented by annexes, where necessary. A good overview of the European legislation can be seen in figure 3.2.

An article may contain several nested sub-levels, each of which can express a distinct legal requirement. To ensure that each requirement identified in this work can be traced to a unique and precise location in the legal text, the methodology preserves the complete supported hierarchy beneath an article. Figure 3.3 presents these possible sub-article levels and their containment relationships.

3.3 Digital Operational Resilience Act

Digital Operational Resilience is becoming more relevant as time progresses. Digital transactions are everywhere and financial services are used with increasing frequency through digital means. This means that information and communication technology disruptions or attacks have the potential to cause financial instability, impacting both individual financial institutions as well as the entirety of the financial system. [20] This means that Digital Operational Resilience is of significant importance for the entire system.

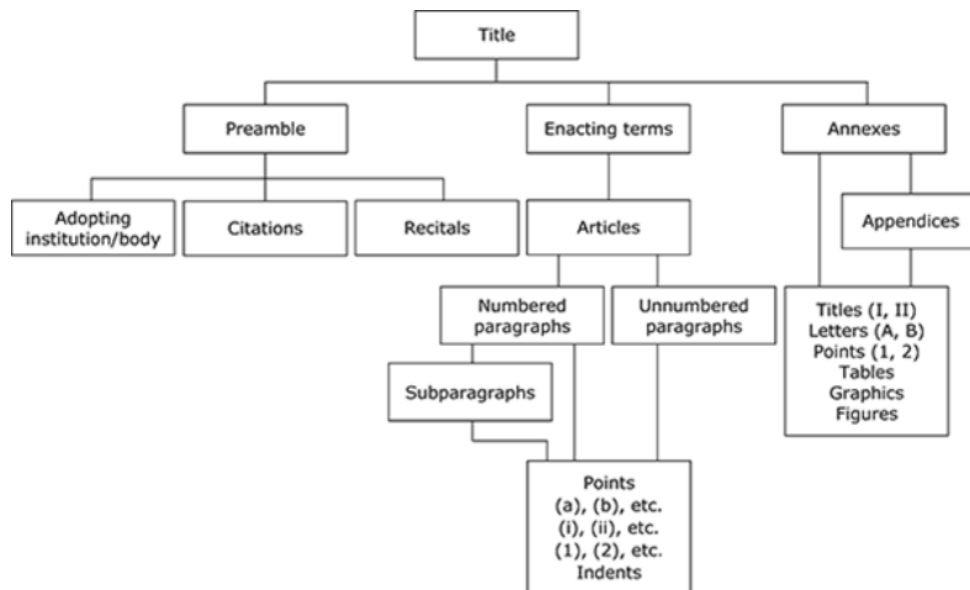


Figure 3.2: Structure of an act [2]

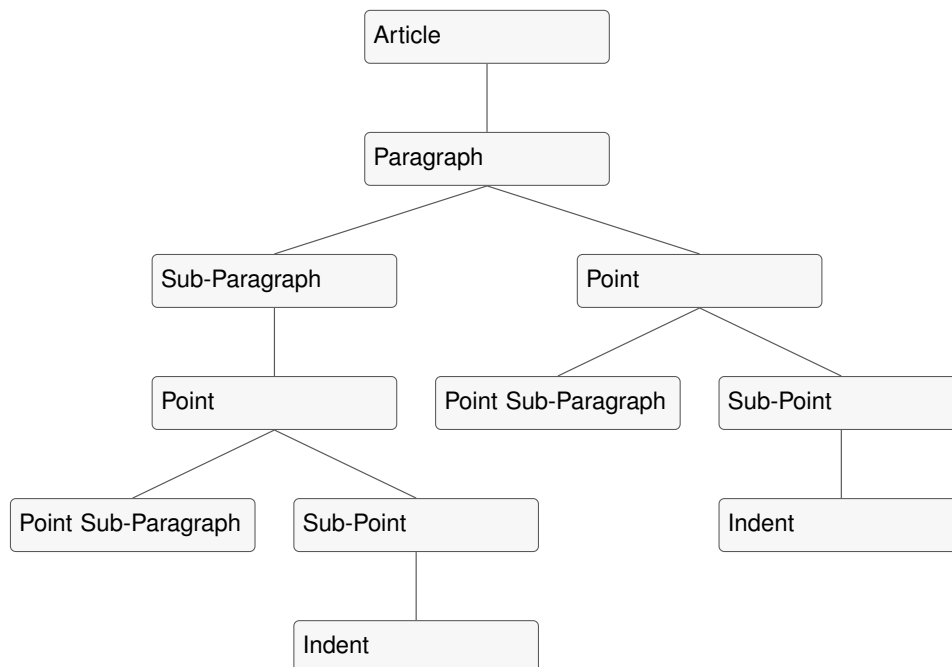


Figure 3.3: Supported containment hierarchy for sub-levels below an Article. Each connecting line denotes immediate containment.

Regulation (EU) 2022/2554 on digital operational resilience for the financial sector, in force since the 17th of January 2025 is a EU regulation primarily focused on ensuring financial entities manage risks arising from the use of Information and Communication Technology (ICT). Due to the scope defined in Article 2, it applies to a wide range of financial entities and aims to create a uniform standard across the EU through the creation of a standardized regulatory framework [20]. This is achieved by delineating precise criteria for ICT risk management that financial institutions must comply with.

1. **ICT Risk Management:** Financial entities must establish and maintain ICT risk management frameworks to identify, detect, respond and recover from digital risks. This includes the creation of governance frameworks and procedures to effectively manage ICT risks.

The requirement placed by DORA for financial institutions to have a strategy to recognize and reduce digital threats goes beyond conventional cybersecurity measures. It promotes the creation of resilient business continuity plans and advanced incident response mechanisms. The implementation of these measures ensures that financial institutions can sustain vital operations in the event of a cyber attack or digital disruption, reducing the impact on financial stability. [20]

2. **Incident Reporting:** DORA requires the establishment of an effective incident management framework for reporting major ICT-related incidents to authorities. This promotes a better understanding of emerging threats, enables a coordinated response and contributes to a shared comprehension of the ICT risk environment. [20]

The purpose of implementing systematic incident reporting following DORA's guidelines is to promote transparency and collective vigilance. By promptly reporting significant ICT-related incidents to relevant authorities, a coordinated approach to managing cyber threats is facilitated, enabling on-time interventions and the dissemination of critical threat intelligence across the sector. This information broadcast enhances the ability of financial institutions to withstand and recover from ICT disruptions, but also bolsters overall stability and protection of the entire financial system. [20]

3. **Digital Operational Resilience Testing:** Testing under DORA comprises both regular digital operational resilience testing and, for financial entities identified by their competent authority, advanced threat-led penetration testing (Threat-Led Penetration Tests (TLPT)). Regular testing is used to assess the institution's preparedness for ICT-related incidents, identify weaknesses in digital operational resilience, and support the remediation of identified deficiencies. Testing is risk-based and includes both technical controls and the processes used to manage and respond to ICT disruptions. [20]

In regard to regular testing frequency, DORA states that "financial entities, other than microenterprises, shall ensure, at least yearly, that appropriate tests are conducted on all ICT systems and applications supporting critical or important functions." Other testing obligations, such as the

testing of backup, restoration, recovery, and exit procedures, are subject to periodic or risk-based testing requirements. [18]

TLPT is an advanced form of testing. It is required at least every three years for the non-micro financial entities identified by the competent authority. These tests should cover several or all critical or important functions of a financial entity and are performed on the relevant live production systems. They are based on the tactics, techniques, and procedures used by realistic threat actors in order to evaluate how well an organisation's resilience measures work against sophisticated cyber threats. [18]

Other types of regular tests are also mentioned, such as vulnerability assessments and scans, open source analyses, network security assessments, gap analyses, physical security reviews, questionnaires and scanning software solutions, source code reviews where feasible, scenario-based tests, compatibility testing, performance testing, end-to-end testing, and penetration testing. [18]

4. **Third-party Risk Management:** Recognizing the growing dependence on third-party service providers, DORA mandates the implementation of rigorous management strategies to address ICT risks that arise from outsourcing agreements in the supply chain, ensuring that financial entities have oversight over the resilience of their critical third-party providers. [20]

DORA's emphasis on the significance of adopting strategy for managing risks by applying the principles of operational resilience across the entire supply chain comes from the realization that a chain is only as strong as its weakest link. The dependence of financial institutions on external service providers (such as the cloud) has led to an increased risk of ICT issues emerging from these given providers. By requiring that financial institutions enforce strict supervision and risk management protocols for their third-party vendors, DORA is guaranteeing that these external collaborators comply with equivalent levels of digital resilience. [20]

5. **Oversight of critical ICT third-party service providers:** In addition to requiring financial entities to manage ICT third-party risk, DORA establishes a Union-level Oversight Framework for critical ICT third-party service providers. The European Supervisory Authorities, acting through their Joint Committee, designate providers as critical where:

- Their failure could have a systemic impact on the stability, continuity, or quality of financial services;
- They support critical or important functions;
- They are difficult to substitute.

For each designated critical provider, a Lead Overseer is appointed from one of the European Supervisory Authorities. The Lead Overseer assesses the provider's ICT risk-management arrangements, including security, availability, continuity, incident management, data portability, testing of ICT systems and controls, and ICT audits. It must also establish an annual oversight plan for the provider.

The Lead Overseer may request relevant information and documentation, conduct investigations and on-site inspections, and issue recommendations concerning identified risks. Where a critical provider does not comply with requests for information, investigations, inspections, or required reports, the Lead Overseer may impose a periodic penalty payment of up to 1% of the provider's average daily worldwide turnover. [18]

6. **Information Sharing:** DORA encourages financial entities to share cyber threat intelligence and other relevant information to enhance collective understanding and defence mechanisms against ICT threats. This collaboration creates a collective intelligence, strengthening the sector's overall ability to withstand ICT threats.

The use of this collective intelligence methodology allows entities to acquire knowledge from the experience of others, exchange optimal methods and work on more efficient strategies to manage digital risks. This collaboration is essential to improve the industry's capacity to adjust to the challenges presented in the current day and age.

It is important to note that this work uses Regulation (EU) 2022/2554 (DORA) as its sole binding legal source for the identification and traceability of regulatory requirements. Regulatory Technical Standards (RTS) and Implementing Technical Standards (ITS) adopted under DORA, including Commission Delegated Regulation (EU) 2024/1774, are outside the legal-analysis scope of this work. This decision limits the methodology to the primary obligations of the regulation and supports a focused assessment of stakeholder-specific regulatory impact.

4

Related Work

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4.1 Automation and AI in Regulatory Analysis and Modelling

The evolution of RE has undergone several transitions over the years, marked by a significant transition pivoting from traditional manual methods to more agile approaches that emphasize increased automation to reduce time, labour and human error. Early efforts in automated RE primarily focus on the analysis phase-identified as the most widely automated phase, according to Umar and Lano [21]-utilizing Natural Language Processing (NLP) technologies such as Part-of-Speech tagging and syntax parsing to generate Unified Modelling Language (UML) models and perform consistency checks [21]. However, these traditional methodologies often struggled with the nuances and ambiguities of real-world legal documents, requiring a shift toward more sophisticated AI solutions [13].

The emergence of GenAI and LLMs represents a fundamental paradigm shift characterized by “emergence” (the ability of models to derive knowledge without explicit programming) and “homogenization”, where a unified foundation model can tackle a wide range of specialized problems [15]. A problem arises when the model does not have sufficient information, limiting answer capabilities and creating a phenomenon called “hallucination”. To address this issue, the Retrieval-Augmented Generation (RAG) model has emerged. This model stores information in databases and retrieves the necessary information to provide it to the LLM when needed, grounding responses in specific enterprise data or legal texts [15]. Unlike NLP, generative LLMs synthesize and create content, allowing RE research to focus on more “cognitively demanding activities”, such as requirements elicitation, validation, and modelling rather than just defect detection and classification, which were dominant in the past [13].

In the regulatory domain, these capabilities have enabled the development of specialized frameworks like MUlti-layered Regulatory Change Identification and Analysis (MURCIA). MURCIA is “an automated approach that leverages recent language models to assist human analysts in analysing regulatory changes.” This approach utilizes LLMs to navigate hundreds of legal provisions and analyse the implications of regulatory changes over time, categorizing impacts across textual, semantic, deontic, and pragmatic layers. Evaluations of this AI-enabled approach demonstrated high efficacy, with an F1 score exceeding 90% for identifying textual changes and text meaning [12].

Additionally, research is also exploring the possibilities of a conversational approach. This approach utilizes a conceptual model interpreter where a modelling task is specified in natural language via a conversational user interface, and an LLM generates model syntax that is automatically rendered with instant feedback. This allows for an iterative process with refinement by specifying a new modelling task. This process can be applied to RE, and also lowers the entry barrier for modelling and design activities [14].

Despite these advancements, a critical gap remains: current studies are often evaluated in controlled laboratory environments with limited integration into the complex, multi-regulatory workflows required in industry [13]. There is a “notable absence of systematic methodologies” [22] that can bridge the abstraction gap between high-level regulatory extraction and the formal consolidation of these insights into an organization’s enterprise architecture.

4.2 LLM Agents and Skills

LLM-based agents extend stand-alone language models by coupling language-based reasoning with planning, memory, tools, and interaction with an execution environment. Rather than only generating responses, an agent can interpret a goal, select actions, invoke external capabilities, observe their results, and adapt its subsequent actions accordingly [23]. This ability is relevant to complex modelling work-

flows, where the work includes information retrieval, evidence processing, structured transformation, validation, and model generation.

Recent literature distinguishes between agents and agent skills. The agent is the higher-level component responsible for interpreting intent, decomposing goals, and coordinating actions. A skill is a bounded and reusable artefact that specifies how a task should be performed under defined conditions [23, 24]. Thus, whereas a tool exposes an atomic capability, such as querying a database or producing a diagram, a skill captures the contextual knowledge required to determine when that capability should be used, how it should be sequenced with other actions, and how its outputs should be checked.

Zhou et al. conceptualise an agent skill as a tuple consisting of a primary instruction document, auxiliary resources, and applicability conditions. The auxiliary resources may include reference material, templates, schemas, executable scripts, or domain artefacts [23]. Xu and Yan similarly describe skills as self-contained packages that may combine a structured instruction file with supporting code, documents, and assets. This packaging allows knowledge to be externalised, inspected, reused, and revised without requiring the underlying model to be retrained [24].

A central architectural principle is progressive disclosure. Skill metadata can be available for matching a task to a relevant capability, while detailed instructions and additional resources are loaded only after the skill is selected [24]. This reduces unnecessary context consumption and enables a larger library of specialised procedures. Skills are also complementary to tool-connectivity mechanisms such as the Model Context Protocol: such mechanisms provide access to external tools and data sources, whereas skills provide the procedural guidance for using them reliably within a workflow [24].

Skills may be authored by domain experts, derived from experience or task traces, retrieved from a repository, composed for more complex workflows, and subsequently revised or retired. However, the literature also identifies open concerns regarding portability, skill selection at scale, composition, testing, provenance, permissions, and security [23, 24]. These concerns are especially relevant where skills operate on legally significant sources or generate architecture artefacts that may influence stakeholder interpretation.

In this context, agent skills provide a suitable mechanism for operationalising a constrained GenAI-assisted methodology. The agent can coordinate the overall workflow, while skills encode bounded procedures for activities such as stakeholder elicitation, legal-text traceability, viewpoint scoping, ArchiMate validation, and cross-artifact reconciliation. Such skills should guide and constrain the use of tools and models, rather than act as an independent source of legal or architectural authority.

4.3 Multi-Agent Debate

Multi-agent debate (MAD) is a collaborative reasoning approach in which multiple large language model agents analyse the same problem, exchange arguments, challenge competing interpretations, and produce a collective conclusion. Unlike a single-agent approach, MAD makes disagreement an explicit part of the reasoning process. Its intended benefits are improved accuracy, reduced individual bias, stronger factual grounding, and more transparent justification of the final output [25, 26].

Oriol et al. [25] identify three main dimensions of MAD design. The first is the participants: beyond debaters, a system may include a moderator or leader, a summarizer, a judge, a verifier, or an editor. Agents may also receive distinct domain backgrounds, positions, or personality traits to encourage diverse reasoning. The second dimension is the interaction mechanism, including the communication topology, turn-taking protocol, and form of exchanged information. Debate may be bilateral or involve several agents; it may be sequential, simultaneous, or hybrid; and agents may exchange complete arguments, summaries, confidence estimates, or factuality assessments. The third dimension is the agreement mechanism, which may rely on collective voting, weighted scoring, averaging, or a judge-based decision.

A central design choice is whether agents formulate their initial positions independently before seeing competing arguments. Simultaneous first-round assessments can reduce ordering effects and preserve genuine differences of interpretation. Later rounds permit agents to rebut, refine, or withdraw their arguments in response to peer feedback. In the requirements-classification study by Oriol et al., a debate between two stance-based agents and a judge improved classification performance over a single-agent baseline. However, the results also show that additional interaction rounds substantially increase token use, execution time, and monetary cost, while yielding only marginal additional improvement. MAD therefore requires a balance between deliberative depth and efficiency.

Gómez et al. [26] present MAD as a structured deliberation process consisting of four phases: initialization, argumentation, refutation and adjustment, and synthesis and consensus. In their design, a moderator defines the objective and rules, specialized agents independently formulate evidence-based proposals, and subsequent exchanges focus on unresolved technical conflicts. The moderator preserves the debate history, controls turn-taking, limits redundancy, and synthesizes the final result. Their findings suggest that structured debate is particularly valuable for complex, strategic, or conflict-laden tasks, whereas a single model may remain more efficient for straightforward retrieval-oriented questions. They also highlight that role and model diversity can reduce premature convergence and groupthink.

The debate panel used in this work applies these principles to the assessment of legal relevance. Its differentiated evaluators provide complementary literal, systemic, and gap-oriented perspectives. The moderator manages structured confrontation and records the basis for consensus or disagreement. Requiring explicit agreement across all perspectives favours conservative, traceable inclusion decisions,

while separately preserving unresolved cases for human review. MAD is therefore justified here to improve accuracy in evaluating relevant requirements against an operational footprint.

4.4 Previous Work

The methodology presented in this paper builds on the value created by a previous dissertation, “Conceptual Modelling of Legal Compliance Requirements - The cases of eIDAS2 and NIS2”. This provided a methodical and structured approach, composed of four main steps:

1. The creation of an Excel Traceability Matrix as an external legal reference;
2. Developing an ArchiMate Index for internal architectural navigation;
3. Connecting both through a hierarchical numbering system;
4. Constructing the ADs according to the defined architectural principles.

4.4.1 Excel Traceability Matrix

Firstly, legislation is broken down into smaller and clearer parts, under the format of an Excel Traceability Matrix. This matrix provides a structured overview of legal text broken down into identifiable units. These units are linked to the ArchiMate AD through a numbering system used consistently throughout the AD all the while serving as a mapper for navigation. The matrix thus acts as an external reference point for navigating legislation. Colours are used to distinguish between different chapters or types of content, improving visual clarity. A sample of a traceability matrix can be observed in table 4.1.

Table 4.1: Sample from the traceability matrix of NIS2 from the previous work. [3]

Chapter	Article	ID1	ID2	ID3	ID4	ID5	ID6	ID7	Text
CHAPTER VI		6							CHAPTER VI - INFORMATION SHARING
CHAPTER VI	Article 29	6	29						Article 29 - Cybersecurity information-sharing arrangements
CHAPTER VI	Article 29	6	29	1					1. Member States shall ensure that entities falling within the scope of this Directive and, where relevant, other entities not falling within the scope of this Directive are able to exchange on a (...)
CHAPTER VI	Article 29	6	29	1	1		a		(a) aims to prevent, detect, respond to or recover from incidents or to mitigate their impact;
CHAPTER VI	Article 29								...
CHAPTER VI	Article 30	6	30						Article 30 - Voluntary notification of relevant information
CHAPTER VI	Article 30								...

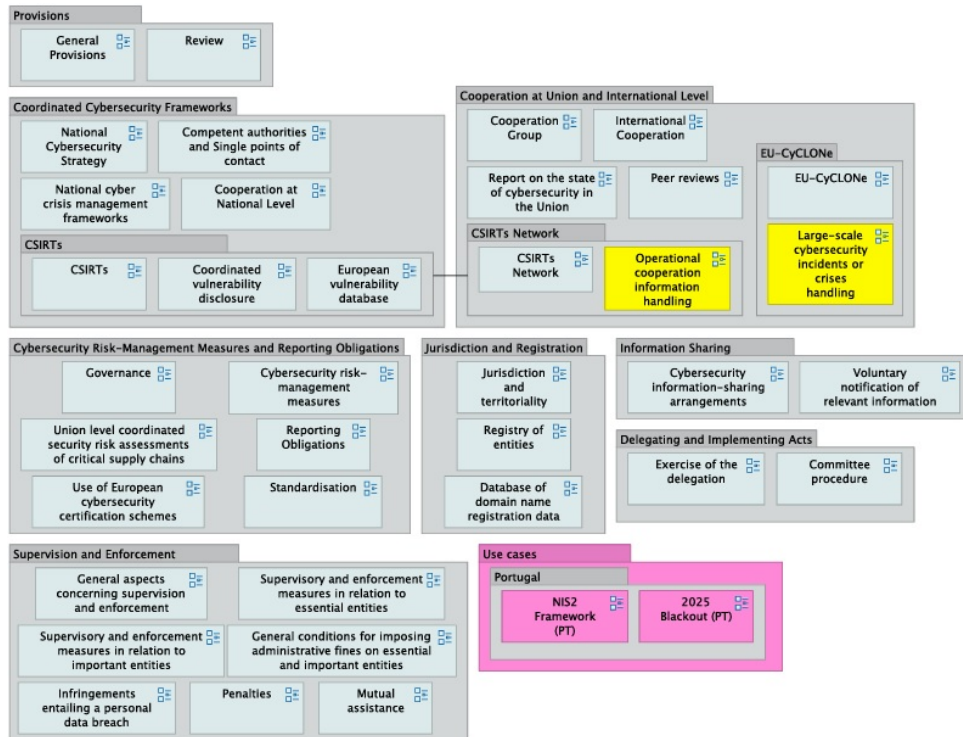


Figure 4.1: ArchiMate NIS2 Index View from the previous work [3]

4.4.2 Archimate Index

Secondly, the ArchiMate Index builds on the traceability matrix to establish an entry point, in ArchiMate, for each AD. This establishes the “Index” view. The “Index” view enables internal navigation between chapters and articles of the legislation and their corresponding architectural representations from a single location [3]. In the “Index” view, each chapter of the legislation is represented as an ArchiMate group, which contains the corresponding ArchiMate views that model each legislation article. Most articles have a view that represents them, however, some were combined into a single view, subdivided across multiple views, or integrated into other related views. These variations are easily understandable due to the numbering system found within the documentation of each view element, which contains an Identifier (ID) that can be traced to the corresponding legal clause using the traceability matrix. Some views do not directly correspond to specific articles but serve either a complementary function or represent a practical scenario. These views are marked in contrasting colours to differentiate them from article-specific views. **Pink** represents practical scenarios and **yellow** depicts complementary views. An example of the index view can be observed below in figure 4.1

4.4.3 Numbering System

The Numbering system is what enables traceability across the legislation, the traceability matrix and the elements in the AD. By providing a simple ID mapping between the AD's elements and the legislation clauses listed in the matrix, it is simple to trace elements to legislative acts. In greater detail, the traceability matrix assigns a unique hierarchical ID to each clause of the legislation. Each ID level corresponds to a structural layer in the legislation. "For instance, the clause "Article 2, (2), (a), (i)" might appear straightforward in the legislation, but in the matrix, it is represented as "1.2.2.1.1.a.i" rather than "1.2.2.a.i". This is because the matrix captures every intermediate level, including numbered lists or grouping structures that are sometimes implicit in the original text. In this case, "1.2.2" represents Chapter I, Article 2, paragraph 2. The additional "1.1" accounts for nested list levels within that paragraph before reaching point (a), and "i" represents the sub-point." [3] Cases where an element is associated with more than one ID exist, meaning that such element is associated with more than one clause or article. Special cases of this include the wildcard, as in "1.1.*", denoting all sub-clauses, and "1.-.1" indicating the absence of a Section.

4.4.4 Constructing the Architecture Descriptions

Lastly, the method finalizes with the development of the ADs for the specified regulation for both NIS2 and eIDAS2. This was done by adopting a top-down approach that transitions from a high-level legislative perspective to a detailed technical modelling. Each piece of legislation underwent a comprehensive chapter-by-chapter and article-by-article analysis to identify elements relevant to the architecture, all the while prioritizing the business layer of ArchiMate to capture how the regulations impact organizational operations, roles and processes.

In order to maintain clarity and interpretive accuracy across complex legal structures, four core principles guided the development of the ADs:

- **Abstraction:** Views are maintained at a level of detail that avoids unnecessary complexity, ensuring stakeholders can focus on relevant legal requirements without becoming overwhelmed;
- **Reuse:** Coherence is achieved by consistently reusing defined legal elements, style types, and architecture patterns, such as event-driven or process-driven designs, throughout both ADs to reduce duplication of effort;
- **Traceability:** A cornerstone of the methodology, the numbering system links every element in the ArchiMate models back to its specific legal clause in the traceability matrix, enabling verification and easier maintenance;

- **Modularity:** The ADs are designed to be independent; each view typically corresponds to a single article or legal unit. This allows stakeholders to understand specific areas in isolation without having to navigate the entire repository.

Each element in the AD is documented with references to official legislative sources. This documentation provides essential context for the independent views, ensuring that important nuances are captured and reducing the need to constantly cross-reference the original legislation or matrix.

To manage the complexity of legal modelling and support iterative refactoring, version control was integrated into the Archi tool using the coArchi2 plugin. This tool integrates Git-based functionality directly into the modelling environment, allowing for the tracking of changes and comparison of modifications.

4.4.5 Results from Application to NIS2 and eIDAS2

The application of the method to the NIS2 and eIDAS2 resulted in ADs that are comprehensive, coherent and navigable representations of both legislations. These are made up of more than a hundred views and over a thousand numbered elements.

The NIS2 AD focuses on operationalizing cybersecurity obligations for essential and important entities, national Computer Security Incident Response Team (CSIRT)s, and European-level cooperation institutions, while the eIDAS2 AD models the expanded legal and technical scope introduced by the revision on the original eIDAS. It covers the legal framework for digital identity, including its means, methods and mechanisms for secure identification, authentication and trust services across the EU.

The viewpoints produced by this work are demonstrable, but they were not formally validated, resorting to GenAI on the issue of validation.

This work explored the use of GenAI to support the validation and documentation of complex ADs, resorting to LLMs such as ChatGPT and an AI-enhanced JArchi integration with LLaMA and Gemini. This exploration does not aim to replace formal validation, but rather position GenAI as a method supporter and stressor. To achieve this, two primary approaches were explored:

- **Unit-Based Review with ChatGPT 5:** In this approach, small “units” of the AD, such as views linked to a single article, were selected and provided to the LLM together with their traceability information and context. This context included ArchiMate XML, relevant legal text, traceability matrices and a list of tasks to perform. ChatGPT 5 successfully captured the essence of views provided in this manner and generated a positive output regarding given tasks. A very accurate clause-to-element table was generated, containing only a minor misalignment. Generated validity assessments also provided useful insights, such as the identification of missing legal provisions, and good feedback on view layout and readability was also offered.

- **AI-enhanced JArchi Integration:** JArchi is a JavaScript-based scripting extension for the Archi tool that allows for the customization of ArchiMate modelling tasks through JavaScript. To exploit its potential with LLMs, two complementary approaches were integrated:
 - **LLaMA 3.1 (Documentation Augmenter):** LLaMA “effectively demonstrated its value as a supportive assistant rather than a substitute for the expertise of the architect” [3]. This is due to post-editing being necessary on the generated output, such as “aligning with the legal provisions, filtering unnecessary information, and correcting misalignments or digressions” [3].
 - **Gemini (Validation Critique):** Gemini was used to generate a validation “critique” of a selected view. Overall, good comments were generated suggesting more critical improvements, identifying possible omission and layering inconsistencies directly within ArchiMate.

It is important to note that this is only a very short summary of an entire masters thesis, and only the points considered to be the most relevant for this work were covered. Many more lower level details can be found on both methodology and results in the thesis “Conceptual Modelling of Legal Compliance Requirements - The cases of eIDAS2 and NIS2”.

5

Problem Analysis

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The main problem this work attempts to solve is informing someone of what actually changes, operationally speaking, due to legislation. This is an important task, since a lot of people affected by legislation know nothing about how they are being affected by it. The objective is to produce artifacts that ease the understanding of how a legislative act affects that stakeholder, specifically. Stakeholder-Specific viewpoints are thus important to achieve this, given that such particular Views must be produced through the viewpoint of the stakeholder. GenAI can be a powerful tool to assist in correctly producing these viewpoints in a timely manner.

The following chapter covers the identified limitations on the work this paper succeeds, as well as the previously identified problems. Through-out the chapter, requirements arising from the analysis of the problem will be identified. It is relevant to note the different actors that take a part in the process: The

analyst or enterprise architect that utilizes the method is identified in the following text as “user”, and the person whose viewpoint is being identified is the “stakeholder”.

5.1 Limitations on Previous Work

The existing methodology provides a structured approach for translating legal requirements into ArchiMate architecture descriptions through traceability matrices, hierarchical identifiers, and modular views. However, important limitations affect the strength of its conclusions and its practical scalability. First, the manual construction of large-scale architecture descriptions requires substantial time and creates a residual risk of omissions, inconsistent interpretations, and modelling errors. Secondly, the methodology lacks external validation from legal experts and intended stakeholders, since its demonstrations are illustrative rather than field-tested. The following subsections examine these limitations in greater detail.

5.1.1 Time and error risk in large-scale manual modelling

The methodology faces a scalability limitation arising from the manual construction of more than one hundred views and over one thousand numbered elements. Creating, interpreting, and maintaining this volume of legal-to-ArchiMate mappings requires substantial time and sustained expert attention. As model size increases, the risk of omitted clauses, inconsistent interpretations, incorrect relationships, duplicated elements, and documentation mismatches also increases. Even when traceability identifiers are used, they do not eliminate the need to verify whether each legal provision has been represented faithfully and consistently across the full architecture description. The approach is therefore valuable for structured legal modelling, but its time cost and residual error risk may limit its practical adoption unless supported by systematic review procedures, automation, and independent quality checks.

This work leverages on the previously explored approach and makes an attempt at partially automating it through the use of GenAI, with pre-defined approval gates for human validation. It is also pertinent to note that the focus changes from representing an entire regulation and its intricacies to representing only what clearly affects the operational footprint of a single stakeholder. To communicate a more global vision of the regulation, in a manner similar to the previous work, an “overview” View is generated. The objective is to let the stakeholder know there is more to the regulation than what produces a direct impact on one’s reality.

5.1.2 External validation

The previous study did not include a formal external-validation phase. The same holds true about this work, due to time constraints. Its practical scenarios were intended primarily to demonstrate how the

methodology could be applied to construct traceable architecture descriptions, rather than to provide stakeholder-based validation of the resulting models.

Accordingly, the results provide evidence of the methodology’s feasibility and potential usefulness, while leaving its interpretation and practical use by legal practitioners, enterprise architects, and regulated organisations open for further study. The AI-assisted reviews offered exploratory feedback during the work, but do not constitute independent legal review or stakeholder validation. External assessment of legal accuracy, usability, and real-world effectiveness therefore remain a relevant avenue for subsequent research.

5.2 Stakeholder-Specific Viewpoints

The legal and operational scope of a normative act is inherently broad. A single regulation may affect multiple organisational units, systems, processes, and governance mechanisms. An architectural representation intended to capture this scope would necessarily encompass many elements that are not relevant to the daily work or concerns of a particular stakeholder.

This creates a practical communication problem. A regulation-wide representation may be valuable from a legal or enterprise perspective, but it can be difficult for an individual stakeholder to determine which aspects affect their own responsibilities [27]. A back-end developer may need to understand how regulatory requirements constrain code changes, testing, deployment, monitoring, and incident escalation. In contrast, a team leader may be concerned with approval responsibilities, delivery dependencies, risk ownership, and resource allocation. Presenting both stakeholders with the same broad representation risks obscuring these distinct concerns with irrelevant detail [28].

ISO/IEC/IEEE 42010 addresses this problem through architecture viewpoints and views. A viewpoint establishes the conventions for constructing and interpreting a representation for specified stakeholders and concerns, while a view applies those conventions to present the relevant part of an architecture description [1]. ArchiMate similarly supports the creation of stakeholder-oriented views that select and relate the concepts needed to address a particular concern [?].

The value of a viewpoint lies in making the intended purpose of an architectural representation explicit. It identifies the stakeholders and concerns to be addressed and establishes the concepts, relationships, and presentation conventions through which those concerns are represented [?, 1]. A viewpoint therefore supports the construction of views that are meaningful to a particular audience, rather than assuming that one universal representation is equally useful to all stakeholders. In order to be meaningful, they must be understood by the person they are meant to inform. This clearly states a requirement the method must fulfil, denoted as **R1**: “Artefacts are comprehensible by the stakeholder”.

The methodology therefore constructs stakeholder-specific viewpoints around the stakeholder’s doc-

umented responsibilities, dependencies, and concerns. Each viewpoint specifies how the regulatory and operational context relevant to those concerns is to be represented, and guides the creation of corresponding ArchiMate views, which support regulatory understanding and discussion.

5.3 Legal to Operational Relevance Gap

European legislation commonly formulates duties at the level of regulated entities, their management bodies, or other legally defined roles. It does not normally specify how an individual employee should perform their daily work. For example, DORA assigns responsibilities for ICT risk management to financial entities and their management bodies, rather than to roles within an organization [18]. The practical effect of a legal requirement on an individual stakeholder must therefore be established through the organisational and operational context in which that stakeholder works.

The legal-to-operational relevance gap is primarily a mismatch of abstraction, terminology, and perspective. Legislation expresses normative expectations through duties, conditions, accountability, and legally defined entities or roles. Operational practice is instead described through work activities, decisions, processes, systems, information flows, and artefacts. Relating the two requires establishing how a legally expressed requirement manifests within a particular organisational context and how that context connects to a stakeholder's work. These connections are rarely stated explicitly in either the legal text or the stakeholder's operational description, which makes regulatory requirements engineering a necessary interpretative step [29].

5.4 Controlled Use of Generative AI

The legal-to-operational relevance gap makes GenAI a potentially useful analytical aid. The process requires the examination of extensive legal text, the structuring of stakeholder-provided operational information, and the comparison of concepts expressed in substantially different language. GenAI can assist with these labour-intensive activities by extracting candidate facts, organising information, identifying potential relationships, and producing structured representations for review. Such uses align with emerging research on AI-supported regulatory requirements analysis [12].

However, generated output cannot itself be regarded as legal, operational, or architectural evidence. GenAI systems may produce plausible but erroneous statements, including content that deviates from supplied sources or presents unsupported conclusions with confidence. National Institute of Standards and Technology (NIST) identifies this risk as confabulation, commonly referred to as hallucination [30]. In the present context, this could result in a fabricated legal reference, an unsupported claim that a requirement applies to a stakeholder, or an architectural relationship that is based on general domain knowledge

rather than documented organisational evidence. These can be mitigated by implementing mandatory human approval-gates, meant to validate the produced output (**R2**: “Human approval required”).

To add to this issue, LLMs are also non-deterministic. The same input will never produce the same exact output twice. In the scope of the current work, this can be an issue when identifying what requirements enforced by a regulation apply to the operational footprint of a stakeholder, specially when dealing with a requirement that applies in a more indirect manner. These eventual divergences can be mitigated through the use of an agent debate panel, previously discussed in 4.3 and identify a new requirement **R3**: “Produced results must be stable”.

The use of GenAI must therefore be controlled through explicit methodological boundaries. Legal statements must remain traceable to a verified legal source (**R4**: “Traceability to the legal source”), while claims regarding stakeholder responsibilities, activities, and dependencies must remain traceable to documented operational evidence (**R5**: “Traceability to operational evidence”) . Where the available sources do not substantiate a relationship, the methodology must not allow the system to infer one (**R6**: Nothing is inferred). GenAI should support the identification and structuring of candidate information, rather than independently determining legal applicability, operational relevance, or organisational compliance.

Human involvement is consequently a substantive validation activity rather than a final confirmation step. A human reviewer must be able to inspect the evidence and rationale behind each material conclusion, reject unsupported output, and explicitly approve the information used in subsequent stages. This preserves accountability for interpretative decisions while allowing GenAI to reduce the effort required to produce evidence-traceable stakeholder viewpoints. The methodology therefore employs GenAI as a controlled analytical aid, not as an autonomous compliance assessor.

6

Methodology

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This chapter discusses the evolutions of the pipeline against the baseline methodology and covers the methodology itself in detail. Using GenAI as a method implementation support tool accelerates development substantially, but sheds light on some issues that need to be addressed in order for GenAI to be helpful instead of hindering progress. These issues are discussed in the 6.1 section. Afterwards, each step of the methodology is covered in detail during section 6.2, with a detailed explanation of the task the LLM accompanying the user performs. Throughout the process, multiple confirmation gates are presented to the user to allow as much control of the output as possible.

6.1 Method Evolution

During development and testing of the methodology, specific pipeline stages presented significant shortcomings that demanded resolution. These phases were namely the “Regulatory Relevance Analysis”

and all the steps that materialized Views (“Contextualized Regulation Overview Baseline” and “View Generation”).

Regulatory relevance analysis proved difficult due to the relative interpretation of legal text in conjunction to the non-deterministic nature of LLMs. Sometimes agents would identify some laws as being relevant to the designated stakeholder and sometimes they would not.

View materialization required two distinct hurdles be surpassed. Due to LLMs having no real spatial awareness, produced Archimate diagrams were cluttered and relations crossed each other multiple times as well as crossing other elements. This resulted in an unreadable view. Therefore, spatial organization and positioning of elements was made agnostic to the LLM. It can’t fail at a task that is not within scope. Besides view organization, GenAI also struggled to develop the Extensible Markup Language (XML) properly. The XML behind an archi model is a very extensive document. If an XML is very deeply nested, the LLM may “forget” the exact opening tag by the time they need to close it, resulting in malformed syntax. In addition to this, the verbosity of XML increases the generation length, which in turn increases the probability that the model will “hallucinate” or make a structural error somewhere during generation. Due to this, most of the times, the generated Archimate file would not even be syntactically valid .

6.1.1 Excel Traceability Matrix

The Excel Traceability Matrix evolved from a manual process to a fully automated one. This means that the time invested to get the Matrix ready diminished from two hours all the way down to a few seconds. A small program was created using C# that analyses the HyperText Markup Language (HTML) from the target legislation, and outputs the Traceability Matrix without any human input.

Besides automation for the sake of efficiency, the traceability matrix also suffered a complete re-design to improve readability. As can be observed on the extract from the baseline Traceability Matrix on 4.1, the ID system is not intuitive. This is because the different IDs can mean any of the sub-article levels, which are not explained anywhere in the matrix. Additionally, of all the groupings referred in section 3.2, only Chapters are mentioned. To address this, every possible grouping that a legislative text may have exists as an ID column. An excerpt of the new traceability matrix can be observed in table 6.1.

Table 6.1: Sample from the traceability matrix of DORA generated by the traceability matrix creator program

Part	Title	Chapter	Section	Article	Paragraph	Sub-Paragraph	Point	Point Sub-Paragraph	Sub-Point	Indent	Text
I											General provisions
I				1							Subject matter
I				1	1						In order to achieve a high common level of digital operational resi
I				1	1		a				requirements applicable to financial entities in relation to:
I				1	1		a		i		information and communication technology (ICT) risk managemer
I				1	1		a		ii		reporting of major ICT-related incidents and notifying, on a volunt
I				1	1		a		iii		reporting of major operational or security payment-related incide
I				1	1		a		iv		digital operational resilience testing;
I				1	1		a		v		information and intelligence sharing in relation to cyber threats ar
I				1	1		a		vi		measures for the sound management of ICT third-party risk;
I				1	1		b				requirements in relation to the contractual arrangements conclud
I				1	1		c				rules for the establishment and conduct of the Oversight Framewo
I				1	1		d				rules on cooperation among competent authorities, and rules on s
I				1	2						In relation to financial entities identified as essential or important
I				1	3						This Regulation is without prejudice to the responsibility of Memb

6.1.2 Diagram as Code

Using “Diagram as Code” made the LLM agnostic to both the verbosity of the XML syntax as well as the positioning of elements. This approach already proved to achieve very good results. Examples of this are Mermaid and LikeC4, with the latter being a closer relative to ArchiMate, built around the concept of an architectural model and not independent visual drawings. Archimate, however, does not fit into this category, since it is a standardized modelling language, not a diagramming method. The approach can, however, be applied to Archimate through the usage of pyArchiMate. This way, the LLM only has to write python code responsible for the generation of the Archimate XML. This never fails syntactically, element positioning is made simpler to the LLM and the model only has to handle a program that, compared to the resulting XML, is a fraction of the size.

6.1.3 Legal Text Interpretation

The relativity of interpreting legal text was mitigated through the use of an agentive debate panel, made up of a moderator and three legislative evaluators. Each evaluator looks at the legislation through a different lens, and confronts it against the stakeholder’s operational footprint, taking note of what is deemed relevant for the stakeholder. The lenses in question are the literal (identifies requirements that explicitly match the stakeholder’s documented responsibilities), systemic (takes into account dependencies between systems, teams and processes that may make a legal requirement relevant) and gap (looks for operational weaknesses and missing controls where the law may expose an unresolved gap). The panel then debates on what should be included in the final report, and the moderator collects the unanimous decisions as well as the ones that did not pass, but had some votes in favour. Everything decided is subject to user approval before being materialized to file. The step-by-step operation of the panel is detailed in table 6.2

To test if the panel did in-fact cause output to be consistent through-out different iterations (thus confirming R3), the exact same input was supplied to the agent responsible for regulatory analysis,

Table 6.2: Order of agentive debate panel

Step	Action
1	Launch the Literal, Systemic, and Gap Evaluator agents concurrently for independent regulatory evaluations.
2	The Moderator compiles the evaluators' initial proposal matrix.
3	The Moderator presents peer proposals and re-invokes the evaluators for multi-turn debate.
4	Retain only the articles on which all three evaluators explicitly agree.
5	The Moderator synthesizes the operational justifications and exhaustive statutory requirements.
6	Obtain user approval, generate the traceability matrix, and validate it.
7	Resolve each in-scope duty to its exact canonical legal-text identifier.
8	Build the bidirectional stakeholder–regulation Coverage Ledger.
9	Creates approved coverage relationships.
10	Map the stakeholder, employing organization, and law-defined legal-role group.
11	Check if user approves Relevant Articles, Omitted Articles, Traceability Matrix, Legal Text IDs, Coverage Resolution and Stakeholder Legal Role
12	Persist the approved report as 5_regulatory_relevance.md.

without any previous context. Only the artifacts materialized from the previous steps were used.

Run	Unanimously Selected Articles	Mixed Selection Articles
1	6–14, 17–19, 24–25	28
2	7–13, 17, 24–25	5, 6, 14, 18, 19, 28

Table 6.3: Regulatory Relevance Analysis distinct runs

The unanimous article selections varied, but the union of unanimous and mixed selections differed by a single article, as is observable in table 6.3. Between both executions, only Article 5 was absent from one run and present (as a mixed inclusion) in another. Every other Article was presented to the user in the suggested regulatory relevance report. This indicates that the panel usually presents a consistent list of articles to the user, with varying degrees of certainty.

6.2 Method Pipeline

The methodology goes through 12 stages, ranging from the initial prompt optimization to the delivery of a fully informative, non-gap accusing model. Multiple agents work throughout every step, making use of a plethora of skills that provide the necessary knowledge basis and restrict LLM behaviour. A full overview of the pipeline is represented in appendix A. Throughout all the stages, the LLM questions the user when it feels it does not have all the necessary information (to prevent hallucination, bolstering **R6**), and before any decision is locked in, ensuring human approval through-out the process (**R2**).

6.2.1 Prompt Optimization

The first step of the pipeline is “Prompt Optimization”. It’s main goal is to achieve the best possible prompt before starting the pipeline. An agent takes the user input, eliminates ambiguity and preserving the original objective, rewrites it using imperative commands and boundary rules. Most importantly, the agent inserts bracketed placeholders for any required fields for which it lacks the necessary information (R6). This ensures a good starting point for the methodology, and serves as a guardrail for the user to provide all the necessary information. When locked in, the prompt produced is written to file, to keep every decision materialized throughout the pipeline. Used agent and skill are referenced in 6.4.

Agent	Used skills	Artifact produced
0-prompt-maker	prompt-maker	1_optimized_prompt.md

Table 6.4: Prompt optimization step.

6.2.2 Interview Design

The second stage of the pipeline is the “Interview Design” step. This step is responsible for the elaboration of an interview designed with the purpose of uncovering the stakeholder’s operational footprint R5. It does this by preparing a jargon-free interview script taking into account the stakeholder that is being interviewed. The interview guide does not expose the underlying regulatory context in order to not produce leading questions. The agent elaborates questions on what the stakeholder builds, changes, uses, reads writes, receives, produces and connects. Questions are grouped into six categories:

- **Core Mission:** Establishes the primary business value and top-level capabilities.
- **Workflow and Processes:** Extracts the sequential business processes and actions.
- **Critical Assets:** Uncovers the software components, systems, and passive data objects.
- **External Dependencies:** Identifies third-party risks, external roles, and integrations.
- **Failure and Resilience:** Extracts disaster recovery paths, physical nodes, and single points of failure.
- **Governance and Ownership:** Identifies the active structure actors, roles, and decision-makers.

For each group of questions, there is a main question, nudges for said question, alternative angles and fallbacks in case the answer is inconclusive. Used resources are present in 6.5. The user must then conduct the interview with the stakeholder and record the responses in the interview script.

Agent	Used skills	Artifact produced
1-interview-designer	regulation-blind-operational-discovery	2_interview.md

Table 6.5: Interview design step.

6.2.3 Context Discovery

“Context Discovery” is the third step in the pipeline. An agent analyses the stakeholder interview responses to extract context regarding their day-to-day operations, processes, and structural responsibilities, as well as dependencies and friction points. Facts are identified with an OP-* ID as well as relationships. Facts are linked to the supporting interview evidence. Explicitly stated relationships are recorded as REL-OP-* entries, documenting their source and target, direction, meaning, and proposed ArchiMate relationship type. This makes traceability to the operational evidence possible (**R5**) Mere co-occurrence of two items in an interview does not establish a relationship. The resulting operational-footprint report captures tasks, assets and data, dependencies, approval gates, incident interfaces, and friction points. The user must approve the extracted facts, relationships, and gaps separately before the report is saved. This provides the evidence-backed operational baseline for later stakeholder classification, regulatory relevance analysis, and viewpoint design. Used resources for this step are present in 6.6.

Agent	Used skills	Artifact produced
2-operational-footprint-extractor	extract-operational-footprint evidence-grounded-relationship-ledger	3_operational_footprint.md

Table 6.6: Context discovery step.

6.2.4 Stakeholder Concerns

The following step down the pipeline is “Stakeholder and Concern Identification”. An agent analyses the operational-footprint report to determine the stakeholder’s organisational class, such as junior, senior, or management, based solely on their documented responsibilities and level of decision-making. The agent also identifies every distinct friction point, bottleneck, operational gap, and concern expressed in the footprint. Each concern is assigned a CON-* ID (**R5**) and, where possible, retains the originating OP-* ID. Concerns are not grouped or summarised: each distinct issue remains separately traceable to its operational evidence.

The resulting stakeholder-classification and concern audit is reviewed through separate user approvals for the seniority classification and the complete concern list. Once approved, it provides the stakeholder perspective and evidence base used by later regulatory relevance analysis and viewpoint scoping.

Agent	Used skills	Artifact produced
3-stakeholder-classifier	extract-stakeholder-class-and-concerns	4_stakeholder_classification.md

Table 6.7: Stakeholder concerns identification step.

6.2.5 Regulatory Relevance Analysis

“Regulatory Relevance Analysis” is the fifth and one of the most detailed steps in the pipeline. A panel of three agents evaluates the stakeholder’s operational footprint, classification, and concerns against the selected EU legislation (**R3**). As previously mentioned, each agent focuses on distinct aspects of the legislation. What follows is a detailed walk-through of the “Regulatory Relevance Analysis” step, described in 6.2.

Agent	Used skill	Artifact produced
4-eu-legislation-identifier		
4a-eu-legislation-literal-evaluator	legislation-relevance-evaluator	
4b-eu-legislation-systemic-evaluator	legislation-relevance-evaluator	
4c-eu-legislation-gap-evaluator	legislation-relevance-evaluator	
4d-regulatory-consensus-moderator	regulatory-consensus-moderator legal-text-anchor-resolver stakeholder-footprint-coverage evidence-grounded-relationship-ledger stakeholder-legal-role-mapper	5_regulatory_relevance.md

Table 6.8: Regulatory relevance analysis step.

The entry point for this step is the “4-eu-legislation-identifier” agent. It is responsible for launching the evaluator agents, handling their proposals to the moderator agent, enforcing the overall sequence of the step and the user validation gates. The “4d-regulatory-consensus-moderator” carries out the substantive activities of this step and therefore uses the most skills, as shown in 6.8. Using the evaluators’ evidence-backed proposals, it reconciles their conclusions, records each approved atomic legal requirement as a REQ-* record, proves its legal source through canonical legal-text identifiers, and maps the requirement to the stakeholder’s work through COV-* coverage records and REL-COV-* relationships. It also records the stakeholder’s legal position through REL-ROLE-* relationships. Because these are distinct tasks, the workflow delegates them to specialised skills. The process and agent interactions are visible in the flow diagram present in 6.1

Each evaluator independently retrieves the target regulation through LexAPI one article at a time and identifies potentially relevant articles. This is done to ensure that every article is analysed. Each one then explains their relevance through an operational anchor (the specific interview evidence, OP-* or CON-*), a legal bridge (how legal duty connects to that operational fact), and the operational ownership or reality (who performs the control) of the requirement. The evaluators also extract the complete set of requirements, restrictions, conditions, and obligations from each candidate article. Their proposals are then reviewed and challenged through a moderated debate. Only articles unanimously accepted by all

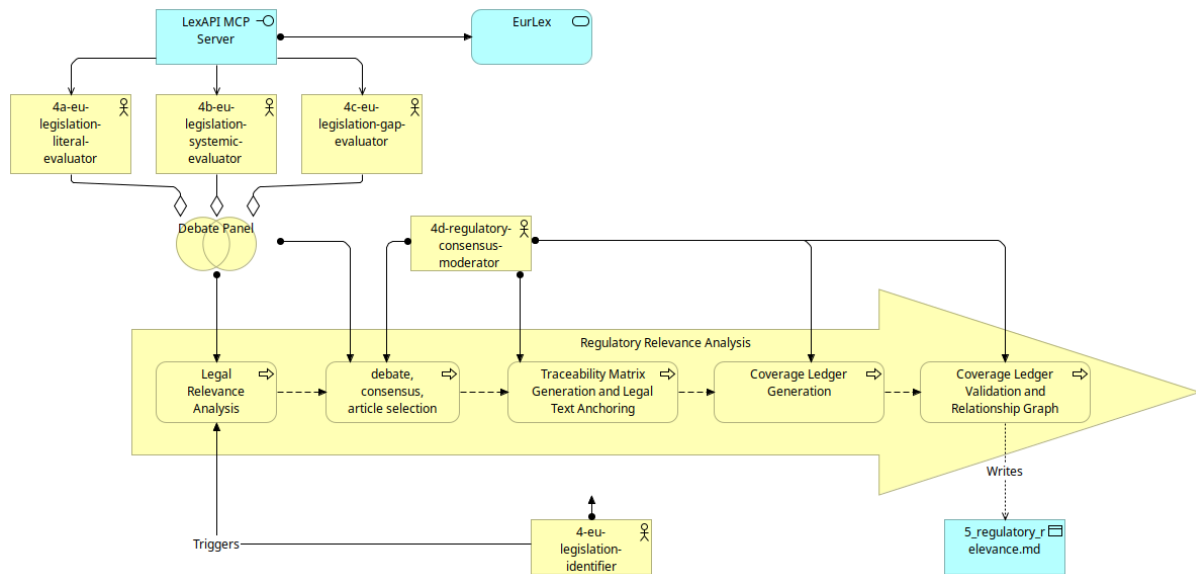


Figure 6.1: Overview of agent interaction in “Regulatory Relevance Analysis” step

three evaluators are retained as relevant. Split decisions are recorded as omitted candidate articles and are presented to the user for validation.

After having fixed the agreed articles, each approved atomic legal duty is recorded as a REQ-* requirement. The process then generates and validates the traceability matrix and links each identified REQ-* requirement to verified canonical legal-text identifiers (**R4**), and builds a Coverage Ledger, in which each entry receives a COV-* identifier. The corresponding approved operational-to-legal relationships are subsequently recorded as REL-COV-* entries (**R5**). A ledger entry provides the following information:

- **Operational Fact Involved:** This is registered using the fact identifier mentioned in 6.2.3.
- **Legal Requirement Affected:** Each legal requirement is given an identifier.
- **Evidence for Connection:** A statement from the interview that confirms an established connection.
- **Stakeholder Requirement Impact :** How this requirement affects the stakeholder.
- **Control Owner:** Ownership is classified as one of the following:
 - **Directly Performed** - The stakeholder performs the relevant control;
 - **Stakeholder-Constrained** - Another control or decision constrains the stakeholder;
 - **Externally Owned** - Another team, supplier or organisation owns the control;
 - **Not Evidenced / Unclear Owner** - Available evidence does not establish an owner.

Afterwards, the ledger is validated so that it can be used to build relationship evidence. To be deemed valid:

- Every operational fact documented from the interview and concern must be connected to either:
 - a legal requirement;
 - a named operational-context stream.
- Every included requirement must have a recorded operational chain, otherwise user clarification is requested.
- The coverage ledger entries become atomic relationship records. These specify the exact operational and legal endpoints, direction, meaning, evidence, and ownership state.

Following coverage resolution, the process maps the stakeholder to their employing organisation and then to the applicable law-defined entity or role group. These remain distinct concepts and are connected through evidence-backed REL-ROLE-* relationships. This prevents the model from incorrectly representing an individual stakeholder as the regulated entity merely because their organisation falls within the regulation's scope.

The resulting regulatory-relevance report is approved through six user gates covering relevant articles, omitted candidates, the legal index, legal-text identifiers, coverage relationships, and legal-role placement. Once approved, it becomes the evidence base for defining the stakeholder-specific viewpoint.

6.2.6 Technology Selection and Overview View Generation

After 6.2.5, the user is prompted to pick a technology. As can be seen in table 6.9, this phase contains three distinct agents, however, only one of them is used, depending on the technology choice made. This is because the view can be generated in three distinct ways:

1. The first option is to generate the .archimate directly, using the agent to write the structured XML syntax the modelling tool (Archi in this instance) can open. For this approach, agent 4e-a is used alongside some specific skills (archimate-xml-generator and archimate-symbol-validator) for correct syntax, model understanding and validation.
2. The second possibility is to generate a view using a different technology, LikeC4. This option was added as an academic experiment to the method. Agent 4e-b achieves better results with LikeC4 when compared to direct .archimate generation. This is due to a key difference, which is that LikeC4 is an Architecture-as-Code (AAC) written in a custom Domain Specific Language (DSL). It was designed from the beginning to be integrated with LLMs, and as a DSL, it is significantly

less verbose. However, the primary scope for this language is software architecture, not enterprise architecture.

3. Lastly, there is also the possibility to pick pyArchimate. pyArchimate is a "Python library for programmatically reading, writing, and manipulating Open Group ArchiMate enterprise architecture models"¹. It is included in the method as an attempt at using the best of both of the previously mentioned options. From the first choice, the preserved benefit is the Archimate output, which is the modelling language with EA in scope. From the second approach, the AAC advantage is kept intact. Therefore, instead of writing a high verbose file, the LLM just needs to write a small python file that is easier to keep in context. The python script written by agent 4e-c then outputs a .archimate file containing the model, justifying the inclusion of the skill archimate-symbol-validator for validation of the generated XML syntax.

After having made a technology choice, the corresponding agent generates the overview view. This view is generated following these steps:

1. **Verify the legal corpus:** Retrieve the consolidated binding text article by article;
2. **Reject out-of-scope sources:** Recitals and generic legal headings cannot become model elements;
3. **Create legal requirement families:** Group binding duties into meaningful control domains;
4. **Anchor legal duties:** Resolve every child requirement to verified Enacting Terms identifiers in the Traceability Matrix. Grouping elements receive the ordered, deduplicated list of their children's anchors;
5. **Model containment and legal relationships:** Nest child requirements inside the corresponding requirement family. Only role-to-family or cross family relationships are drawn;
6. **Add stakeholder context:** Add the approved stakeholder, employing organisation and law-defined entity or role chain to the view to improve readability and establish the legal perspective;
7. **Validate graph integrity:** Every element needs an approved relationship, the stakeholder must be present and every element must be reachable from any other element;
8. **Obtain user approval:** Confirm legal source completeness, obligation domain ,requirement families and the proposed overview map;
9. **Materialize overview view:** The view is materialized to file, to be consumed by latter stages of the pipeline.

¹pyArchimate library is available online in <https://pypi.org/project/pyArchimate/>

The resulting artifact is therefore a technology-specific but semantically invariant baseline for all later stages. It represents the complete regulation-wide legal core through meaningful requirement families and their nested child obligations. The approved stakeholder to organisation to legal-role chain establishes the perspective of the stakeholder regarding that legal core without without treating the individual stakeholder and the regulated entity as one and the same.

Legal citations and verified traceability matrix IDs are retained in model documentation (**R4**). Once fully approved, the overview becomes immutable. Later stages may create stakeholder-specific views, but they must preserve the legal core and stakeholder legal-role chain of the overview view.

Agent	Used skill	Artifact produced
4e-a-archimate-overview-modeler	archimate-xml-generator archimate-symbol-validator	0_regulation_overview.archimate
4e-b-likec4-overview-modeler	likec4-dsl	0_regulation_overview.c4
4e-c-pyarchimate-overview-modeler	pyarchimate-model-generator archimate-symbol-validator	0_regulation_overview.py 0_regulation_overview.archimate
4e-a, 4e-b and 4e-c	regulation-overview-designer legal-text-anchor-resolver stakeholder-legal-role-mapper regulatory-view-structure connected-view-validator	

Table 6.9: Contextualized regulation overview using ArchiMate XML.

6.2.7 Viewpoint Scope

Following the “Regulatory Relevance Analysis”, the “Viewpoint Scope” is responsible for answering the question : “Which parts of the stakeholder’s operational reality and the regulation should be represented, and why?”. No new legal requirements or operational facts are created during this step. The agent responsible uses the skills observable in table 6.10 to filter and organise the approved evidence from earlier steps. It determines the boundary of the viewpoint through chaining requirements to the stakeholder. A legal requirement is included only when it has a recorded operational chain to the stakeholder. Requirements that are applicable to the organisation, without a documented impact on the stakeholder’s work, remain outside the scope (**R6 and R5**).

Agent	Used skill	Artifact produced
5-stakeholder-viewpoint-scooper	viewpoint-scope-definer stakeholder-footprint-coverage evidence-grounded-relationship-ledger regulatory-view-structure connected-view-validator	6_viewpoint_scope.md

Table 6.10: Viewpoint scope and justification step.

The agent groups included material from previous steps (operational footprint, relevant articles, stakeholder classification, coverage and relationship ledger, and regulation overview baseline) into co-

herent operational streams (**R1**). Examples of this, when applied to the DORA regulation from the viewpoint of a developer, are “Protection and Change”, “Resilience Testing”, “Detection and Continuity”, among others streams. Each group is further distinguished as Governance, Administrative, or Technical. Non-regulatory work is retained in a role-specific operational-context stream rather than being incorrectly presented as a legal duty.

Ownership remains an important distinction in this step. A directly performed requirement may later be represented as a stakeholder realisation. A stakeholder-constrained, externally owned or of unclear ownership requirement must be shown through an appropriate dependency, role, approval or constraint.

These groups are also proposed as the future views to be generated. Every candidate view must contain the stakeholder, include only approved relationships, have no isolated elements, and every element must be reachable from any other element. If a proposed view has disconnected parts, it is split into meaningful separate views (**R1**). During the process, the user must approve:

- The defined scope boundary;
- Included requirements;
- Operational streams;
- The connectivity of each proposed view;
- Any eventually required split.

6.2.8 Viewpoint Generation

The “Viewpoint Generation” step converts the scope approved in the previous phase into the formal ArchiMate viewpoint and an authoritative view graph. The viewpoint defines exactly what later artefacts are allowed to represent (strengthens **R3**).

The agent takes the approved scope, Coverage Ledger, relationship evidence, and immutable Regulation Overview baseline. Then, for each view, it defines:

- Purpose and stakeholder perspective.
- Included concerns and operational streams.
- Allowed ArchiMate elements and relationships.
- Requirement-family structure and nesting.
- Visual rules and layout constraints.
- Evidence (**R5**), ownership safeguards (**R6**), and legal-text documentation (**R4**).

The overview view remains the same generated in 6.2.5. It preserves the regulation-wide legal core. Other views will represent the narrower the narrower, stakeholder specific operational scope.

An important output of this step is `7_view_graph.json`, one of the outputs present in table 6.11. It is an authoritative machine-readable graph contract for all later outputs (narrative, slides and model generation). It records:

- Every allowed node and its view membership.
- Every approved REL-* relationship (these include endpoints, direction, meaning, evidence, and visibility).
- Requirement family containment and visual nesting.
- The stakeholder anchor in every view.
- Ownership rules, such as preventing an externally owned requirement from being represented as stakeholder realisation.

The generated viewpoint also perpetuates every restriction defined in the 6.2.7 down to the view level. Before materializing the viewpoint to file, the user must validate the metamodel scope defined by the viewpoint (allowed ArchiMate concepts, relationships and visual rules) as well as the view graph (each view's nodes, relationships, nesting and connectivity result). The resulting viewpoint specification and graph become the fixed contract used to generate the explanatory text, slide diagrams, and final model (**R3**).

Agent	Used skill	Artifact produced
6-viewpoint-materializer	viewpoint-creator	7_viewpoint.md
	viewpoint-materializer	7_view_graph.json
	stakeholder-footprint-coverage	
	evidence-grounded-relationship-ledger	
	regulatory-view-structure	
	connected-view-validator	

Table 6.11: Viewpoint generation step.

6.2.9 Textual View Explanation

The viewpoint and graph can be used to convey information in a multitude of ways. One of them, materialized in this stage, is the `view_explanation`. The agent of this phase, represented in 6.12. It serves two purposes at once: To explain how the regulation affects the daily work of the stakeholder and to explain how to read the corresponding architectural views (**R1**). For this, the agent uses the viewpoint and the view graph created in the previous step as a source of truth, not adding, removing or changing any nodes or relationships.

Agent	Used skill	Artifact produced
7-view-narrative-explainer	view-narrative-explainer stakeholder-footprint-coverage evidence-grounded-relationship-ledger regulatory-view-structure connected-view-validator	1_view_explanation.md

Table 6.12: Textual view explanation step.

The explanation is generated in a per-view basis, starting with the regulation wide overview view. This explanation each stakeholder-specific stream, such as governance, operational administration, or technical work. Then, for each other view, the explanation describes:

- The view's purpose and stakeholder concern;
- Relevant operational tasks, systems and dependencies;
- Applicable requirement families and granular legal requirements;
- Ownership state of each legal effect;
- Relations between elements.

These topics cover the narrative walk-through of the operational and regulatory context. In addition, every stakeholder-specific view contains an explicit "How to read this view" subsection. It explains:

- every visible box, including its ArchiMate type, what it represents, why it is present, and whether it is nested within a requirement family;
- every visible arrow, including its source and target, relationship type, evidence-backed meaning, and why it is represented visibly;
- semantic-only composition or aggregation relationships, which are communicated through nesting rather than an internal arrow;
- the route from the stakeholder to the relevant operational elements, dependencies, external roles, and requirement families;
- external actors, systems, approval gates, and unclear-owner requirements, explaining why they remain outside the stakeholder's direct control;
- the distinction between work directly performed by the stakeholder and work that is constrained, externally owned, or not evidenced.

Before being materialized to file the user must approve, for each view, the view name and the view description. The final artifact also contains a layer-by-layer explanation of the ArchiMate concepts used

across the viewpoint and a complete traceability table. This table connects each Coverage Ledger entry to its operational source, requirement family, legal requirement, ownership state, representing element or external role, and operational meaning.

6.2.10 Markdown Slide Diagrams

The next stage is “Markdown Slide Diagram” generation. The artifact observable in 6.13 is produced by the agent of this step, who turns the approved viewpoint, graph and textual explanation into a presentation-style visual document. The main goal of this step is to present an accessible, condensed visual summary of each approved view (**R1**). The complete view is conveyed by the diagram together with its supporting tables; the diagram itself is not an exact graphical projection of every element and relationship.

Agent	Used skill	Artifact produced
8-informal-diagram-designer	informal-markdown-diagram-designer stakeholder-footprint-coverage evidence-grounded-relationship-ledger regulatory-view-structure connected-view-validator	2_slide_diagrams.md

Table 6.13: Markdown slide diagrams step.

The output begins with the slide for the overview view. Its ASCII diagram is a compressed representation of the immutable legal core. It is followed by one slide for each stakeholder-specific view defined in the viewpoint graph.

Each stakeholder-specific slide contains a condensed ASCII diagram sketch. It shows the stakeholder, the principal operational path, relevant systems or external dependencies, and every requirement family as a labelled group. Accompanying the sketch are two supporting tables:

- **Component Breakdown:** Listing every granular requirement within each requirement family, together with readable legal citations, ownership state, and operational relevance;
- **Relationship Inventory:** Recording every approved relationship by its readable source, meaning, target, ArchiMate type, and ownership.

Lastly, each slide contains an Operational Takeaway in plain language, summarising the direct work, constraints, dependencies, and potential operational or visibility limitations of the stakeholder.

Before being committed to file, the user must approve each ASCII slide individually, including the detailed information retained in the accompanying tables.

6.2.11 View Generation

The view generation is the phase where viewpoints are materialized in the form of views. The architecture model that a modelling tool can open, inspect, reuse and refine. It is not intended to be a finished product, but to be used as a starting point to be improved over iterative efforts. The step consists of only a single agent, despite three distinct agents being present, as shown in table 6.11. This is because the views can be generated in three distinct ways, the same way as in 6.2.6.

Despite the choice made on step 6.2.6, the user is prompted to choose one of these three options. Regardless of the technology option, the agent associated with the choice always requires the same input: The overview view, viewpoint, view graph, textual view explanation and the condensed slides. From these inputs, the graph is the absolute authority while the narrative and the condensed slides help explain and organise the model. The process then works as follows:

1. The agent creates the model elements from the approved graph;
2. Requirement hierarchy is preserved. Each requirement group is a meaningful parent ArchiMate Requirement or Constraint, with its child requirements are visually nested within it (**R1**);
3. Approved relationships are materialized with their exact source, target, direction, and ArchiMate type. A stakeholder may realise a requirement only where ownership is Directly performed (**R5**);
4. The overview view is copied and the stakeholder-specific native views are created. Every view includes the stakeholder and contains no isolated elements. Arrows are only omitted where visual nesting communicates containment;
5. Citations, ownership metadata, and traceability matrix IDs are stored in element descriptions (**R4**);
6. Output is validated in the selected technology:
 - Archimate XML checks structure, folders, diagram objects, connections and coordinates.
 - LikeC4 checks DSL syntax, scoped references and rendered views.
 - pyArchiMate validates relationship legality, executes the generator script.
7. The proposed thematic view layouts are presented to the user for approval before persistence;
8. The generated model is checked against the graph. Missing nodes, extra relationships, broken containments, disconnected views or an altered overview view cause model rejection (**R3**).

Agent	Used skill	Artifact produced
9a-archimate-model-generator	archimate-xml-generator archimate-symbol-validator	3_view_model.archimate
9b-likec4-model-generator	likec4-dsl	3_view_model.c4
9c-pyarchimate-model-generator	pyarchimate-syntax-reference archimate-symbol-validator	3_view_model_pyarchimate.py 3_view_model_pyarchimate.archimate
9a, 9b and 9c	stakeholder-footprint-coverage evidence-grounded-relationship-ledger regulatory-view-structure connected-view-validator	

Table 6.14: View generation using ArchiMate XML.

6.2.12 Cross-Artifact Reconciliation

The last step in the pipeline is “Cross-Artifact Reconciliation”. All necessary knowledge basis are present in table 6.15. This stage is responsible for ensuring that all generated artefacts depict the same information (**R3**). The agent responsible for this phase compares:

- Overview view;
- Viewpoint and the graph;
- Textual explanation;
- Explicative Slides;
- The model;
- Traceability Matrix.

Agent	Used skill	Artifact produced
10-view-cross-artifact-reconciler	view-cross-artifact-reconciler legal-text-anchor-resolver stakeholder-footprint-coverage evidence-grounded-relationship-ledger regulatory-view-structure connected-view-validator	Audit report View Updates

Table 6.15: Cross-artifact reconciliation step.

This is important because each artefact can appear valid on its own while contradicting another. For example, a narrative may describe a developer as constrained by a control while the model incorrectly depicts them as directly realising it, or a requirement could be renamed in slides but retain its original legal anchor in the model. Then validation happens in the following step by step manner:

1. Reads the approved graph as the ground truth;
2. Verifies the overview view against the traceability matrix to check the legal anchors;

3. Checks that each in-scope requirement has an operational chain;
4. Compares the narrative, slides, and model against the approved view elements and relationships;
5. Validates the technology-specific syntax of the model, references, containment, and diagram connections;
6. Produces a discrepancy report if anything differs;
7. Presents the user with a confirmation gate, so that nothing is modified without permission;
8. Applies only the user approved corrections;
9. Reruns the audit.

The gain is assurance: the final model, explanation, and slides can be used by different readers without silently communicating incompatible claims. These artifacts also support each other. If the stakeholder is not proficient in reading Archimate diagrams, condensed slides and explicative text may help with correctly interpreting the views.

7

Results and Demonstration

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This chapter presents the results of the methodology at four complementary levels: stakeholder comprehension, controlled use of generative AI, traceability-artifact automation, and the resulting stakeholder-specific Architecture Descriptions. Rather than treating generated views as the sole outcome, the chapter considers whether the methodology produces representations that stakeholders can understand, whether generative-AI output remains subject to evidence-based and human controls, and whether traceability artifacts can be created and validated consistently.

The first part of the chapter addresses viewpoint clarity and comprehension from the stakeholder perspective, followed by the controlled integration of GenAI for validation and the partial automation of legal traceability artifacts. Together, these sections assess the practical qualities required for the methodology to be useful: comprehensibility to its intended audience, evidence grounding and human accountability, and repeatability in the handling of legal-source records.

The chapter then demonstrates the methodology through its application to DORA and the CSA. These applications examine how documented stakeholder responsibilities, dependencies, and concerns can be related to relevant legal requirements and materialised as stakeholder-specific ADs. For each case, the chapter presents the operational-footprint and concern analysis, regulatory-relevance and traceability results, formal viewpoint specification, and the resulting visual and textual view representations. This makes it possible to examine both the methodology's controls and automation mechanisms and its ability to produce role-relevant architectural representations of regulatory impact.

7.1 Controlled Integration of GenAI for Validation

GenAI is integrated throughout the automated stages of the methodology as one layer of a controlled assurance process. It is not treated as an autonomous authority capable of proving either the factual reality of an organisation's operations or its legal compliance. Instead, the LLM supports the extraction, structuring, comparison, and validation of evidence against the constraints defined by the methodology.

Before an artifact is materialised, the responsible agent checks that its content is grounded in approved source evidence, uses stable identifiers, and satisfies the structural rules of the relevant stage. Where information is ambiguous, incomplete, or unsupported, the process follows a zero-assumption principle: the agent must expose the uncertainty and request user input rather than infer a conclusion. This applies, for example, to operational facts and relationships, stakeholder concerns, legal relevance claims, legal-role mappings, coverage records, and view content.

The methodology combines LLM-assisted validation with deterministic and source-based checks.

- Legal duties (REQ-*) are validated against the traceability matrix and the legal text;
- Relationship and view graphs are checked for valid endpoints, evidence, connectivity, and stakeholder presence;
- Generated native models(Archimate, pyArchimate or LikeC4) are checked through syntax validation, semantic validation, and, where applicable, parsing or round-trip loading.
- The final cross-artifact reconciliation stage verifies that the legal index, coverage ledger, viewpoint graph, textual explanation, slide diagrams, and selected native model remain mutually consistent.

User validation is therefore a required control rather than an exceptional fallback. At each approval gate, the operator may approve, reject, amend, or request further evidence for a proposed result before it is persisted. The resulting validation process provides traceability, internal consistency, evidence grounding, and technical correctness. It does not, however, replace expert legal judgement or independently establish that the organisation is compliant in practice.

7.2 Partial Automation of Traceability Artifacts

The methodology partially automates the creation and validation of traceability artifacts. Here, traceability refers to the ability to move between stakeholder interview evidence, operational facts and concerns, applicable legal duties, and the architectural representations produced later in the pipeline. Automation improves repeatability, reduces transcription errors, and makes the resulting chain of evidence easier to inspect. It does not, however, replace legal interpretation, stakeholder knowledge, or operator approval.

Once the applicable legal articles have been agreed, the agent calls the traceability matrix creator program to generate a traceability matrix. This matrix records the binding enacting terms of the selected consolidated regulation using IDs. It acts as a human and machine readable index of the authoritative legal text, allowing the methodology to resolve a proposed legal duty to its exact legal source. The legal index is validated for its regulation and CELEX identity, expected structure, and the unique resolution of each referenced legal-text identifier.

The automation is deliberately partial. The system can generate proposed records, verify identifier resolution, detect missing anchors, check structural consistency, and preserve provenance across artifacts. It cannot independently establish that an interview statement is factually complete, that an interpretation of a legal provision is correct, or that an organisation complies with the regulation in practice. Consequently, the operator reviews and approves the legal article set, legal anchors, coverage ledger, legal-role mappings, and later reconciled artifacts before they are persisted.

7.3 Architecture Description of DORA

This section presents the application of the methodology to DORA through three stakeholder-specific viewpoints: a Developer viewpoint, an Infrastructure and Cloud Engineer viewpoint and a Team Leader viewpoint. All viewpoints are derived from the same regulatory corpus, but represent different operational responsibilities, dependencies, concerns, and relationships with the regulation.

For each stakeholder, the methodology identifies the relevant operational footprint, maps the applicable legal requirements to the documented work context, and materialises the resulting viewpoint as an Architecture Description, while keeping user corrections only where necessary. The two cases therefore illustrate how the method can retain a common regulation-wide legal basis while producing distinct, role-relevant representations of its operational impact. The LLM chosen for the application of the methodology was GPT-5.6 Terra set to Extra High reasoning, used through the Codex coding partner.

7.3.1 From the Viewpoint of a Bank Developer

The first stakeholder to whom the methodology was applied regarding DORA was a back-end developer of a large banking institution. This stakeholder is familiar with ArchiMate, and does have some understanding of the DORA regulation. Below are covered the most important aspects of some of the generated artifacts, since fully listing them all would be too extensive.

7.3.1.A Operational Context and Analysis Boundary

Through the use of a prompt containing placeholders to be replaced by the user, context and objectives were delineated. The agent from the second step of the pipeline takes the information rich prompt the agent from the first phase helped the user make, and generates an interview, that when conducted with the stakeholder, is used to establish an operational footprint and classify the stakeholder. During these steps, the generated footprint, classification and concerns were accepted without any adjustments from the user. Assessing the interview (which can be found in the Github¹) against the resulting operational footprint, its clear the agent was capable of correctly extracting everyday tasks and dependencies, observable in 7.1.

Dimension	Case profile
Stakeholder and context	Back-end Developer working on an internal Enterprise Data Platform that makes data available to other teams and services.
Core responsibilities	Develops offloads, ETLs, and query services; maintains the internal framework; implements and tests code; creates pull requests; coordinates releases; and monitors production deployments.
Delivery path	Works from functional specifications, performs data transformations and validation, tests in development environments, submits pull requests, opens QUA and PROD change requests, and monitors deployments through Datadog.
Key dependencies	Depends on sufficiently complete specifications, read and write permissions, Team Manager and Solution Architect decisions, Infrastructure Team deployment support, and internal services such as Azure AD, Azure DevOps, Kubernetes, databases, and Kafka.
Decision and incident boundary	The stakeholder implements and tests changes but does not hold final authority for architecture, functional requirements, or production approval. During incidents, the stakeholder participates in escalation and may remediate software errors owned by the responsible development team.
Evidence basis	The operational footprint contains 89 approved, interview-grounded operational records, including tasks, dependencies, approvals, incident interfaces, friction points, and evidence limitations.

Table 7.1: Case profile for the DORA Back-end Developer viewpoint.

¹The Github for this dissertation can be found here: https://github.com/AntonioBrasIST/Architecture_Viewpoints_for_EU_Regulation

7.3.1.B DORA Relevance and Ownership

The extracted operational footprint and concerns are fed into the next step, the regulatory relevance analysis. This step looks through the DORA act searching for legal duties that will directly affect the footprint of the stakeholder, either by affecting a direct task, or by affecting the produced artifacts or some other elements that indirectly connect to the stakeholder. The debate panel explained on section 6.2.5 unanimously retained, after structured debate, articles 7 through 13, 17, 24 and 25. Articles 6, 14, 18 and 19 were omitted from the stakeholder scope due to there being no record of participation on the ICT risk management framework, crisis-communication duty, incident classification duty, or competent-authority reporting responsibilities, respectively.

As a test, Article 28, requiring financial entities to manage ICT third-party risk throughout service lifecycle, was introduced as a relevant portion of legal text that should be listed. This served as a test to check how manual interference would affect the end results and resulting regulatory relevance report, as well as introducing a requirement group that was correctly kept out of scope by the agent panel. The report successfully included all articles approved within the case scope (including 28), and listed every requirement of each article alongside the corresponding traceability matrix ID. It is important to highlight that ownership is being taken into account when mapping requirements to the operational evidence, so that they can be reflected during view artifact generation. The distinct ownership states can be seen on table 7.2.

Ownership state	Interpretation
Directly performed	The documented maintenance work is represented as directly realizing the relevant DORA requirement.
Stakeholder-constrained	The requirement constrains the developer's work, but the developer is not presented as solely responsible for processing capacity.
Externally owned	Access control is relevant to the developer's work but is represented as an externally owned dependency rather than a developer-implemented control.
Not evidenced / unclear owner	The requirement affects the platform, but the interview did not establish which role or team owns its implementation.

Table 7.2: Representative DORA coverage records by ownership state. The table demonstrates traceability, not organisational compliance.

Following coverage resolution, pyArchimate was selected as the technology to generate the overview View. The resulting view grouped some of the requirements DORA mandates into 14 requirement families, observable in figure 7.1. This provides a solid overview of the entire regulation, but not a complete, as will be observed further in 7.3.2.B.

The viewpoint scope was limited to the developer's evidenced platform work and the DORA duties that affect it. The resulting boundary includes framework maintenance, platform development, data offload work, testing and developer readiness, controlled releases, production monitoring, incident interfaces, and the user added Microsoft-cloud dependency. Everything else was excluded from direct

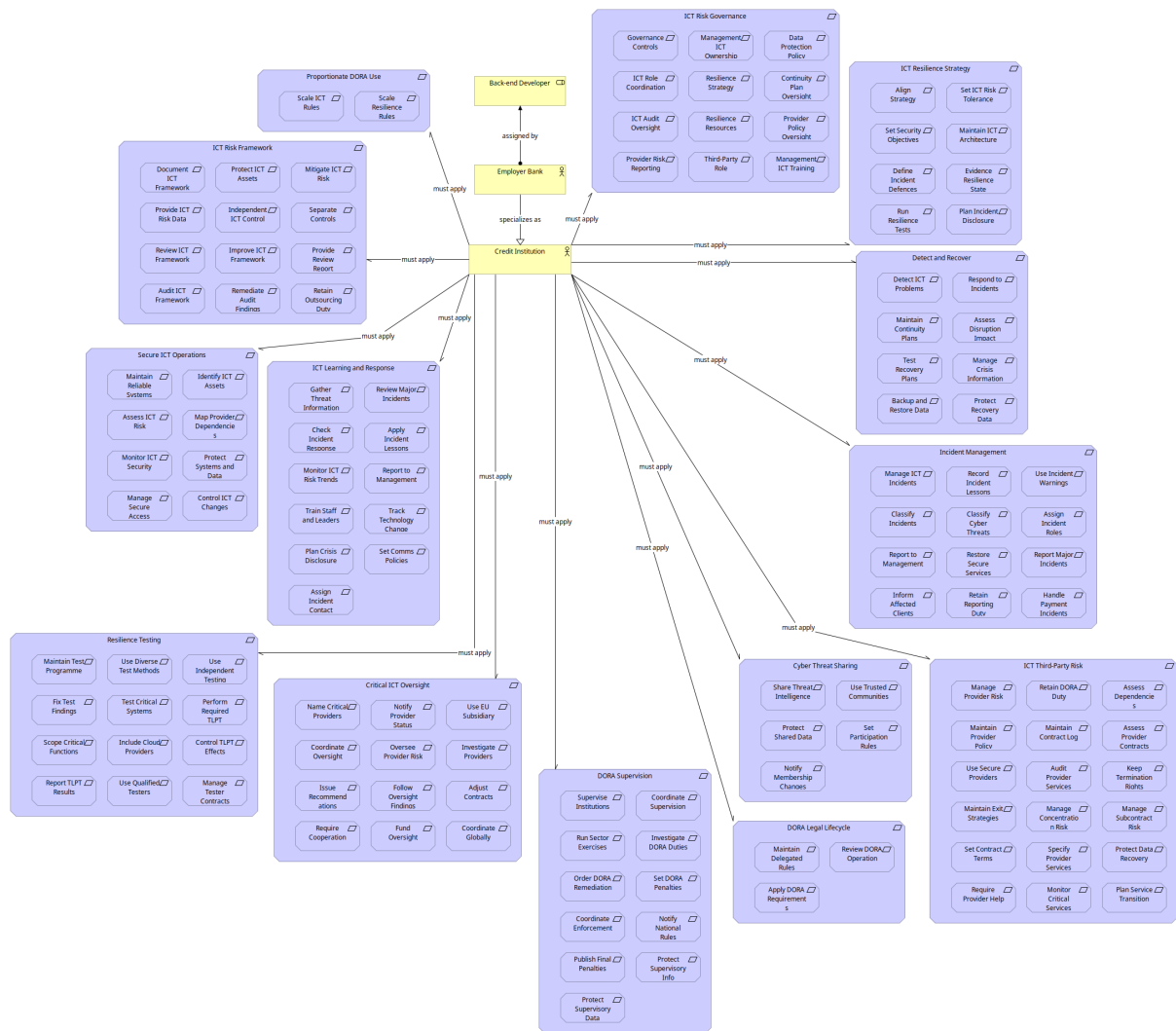


Figure 7.1: DORA requirements overview

stakeholder responsibility where the interview did not support such an attribution. These matters remain represented either as external dependencies, unclear ownership boundaries, or contextual information.

The scope was then materialised into a formal ArchiMate viewpoint. Rather than producing a single large diagram, the viewpoint separates the stakeholder-relevant regulatory impact into six connected thematic views: ICT Asset Governance, Protection and Change, Detection and Continuity, Incident Learning, Resilience Testing, and Microsoft Cloud Risk. A separate regulation overview view preserves the wider DORA legal context. This overview does not represent daily work or make a compliance claim. Its purpose is to establish the legal context in which the stakeholder-specific views are interpreted.

The viewpoint has a designing purpose: It makes approved DORA constraints, ownership states, and operational relationships usable in the resulting AD without asserting enterprise compliance. Its primary concerns are:

- Cloud service provider contract risk;
- Change, testing, and incident interfaces;
- Platform, monitoring, continuity, recovery, and remediation constraints.

These concerns are formalised into six connected thematic views, using nested requirement families and evidence-backed ArchiMate relationships.

7.3.1.C Resulting Architecture Description

The approved viewpoint was materialised as a pyArchimate AD, preserving the approved operational evidence, legal requirements, ownership states, and ArchiMate relationships.

The resulting AD contains a regulation overview view and six stakeholder-specific thematic views: ICT Asset Governance, Protection and Change, Detection and Continuity, Incident Learning, Resilience Testing, and Microsoft Cloud Risk.

The Resilience Testing view, shown in Figure 7.2, represents testing and developer readiness activities, pull-request creation, and the separate Technical Reviewer role. It demonstrates that testing duties directly connected to the developer are represented differently from review activities whose ownership was not established by the interview evidence. It also shows the requirements that constrain the developer, as well as the realized requirements. Every view can be observed in the Github¹.

It is also of interest to look at the generated view for Microsoft Cloud Risk, depicted in 7.3. While the view may look weak at a first glance, it is important to note that operational queues given by the stakeholder during the interview make no mention of a contract with Microsoft, only implications. Additionally, these concerns do not affect the footprint of the stakeholder in any way. The view should not exist in the first place, and would not exist had it not been force-included by the user during the requirement elicitation phase. Therefore, the view was generated in accordance with existing operational evidence

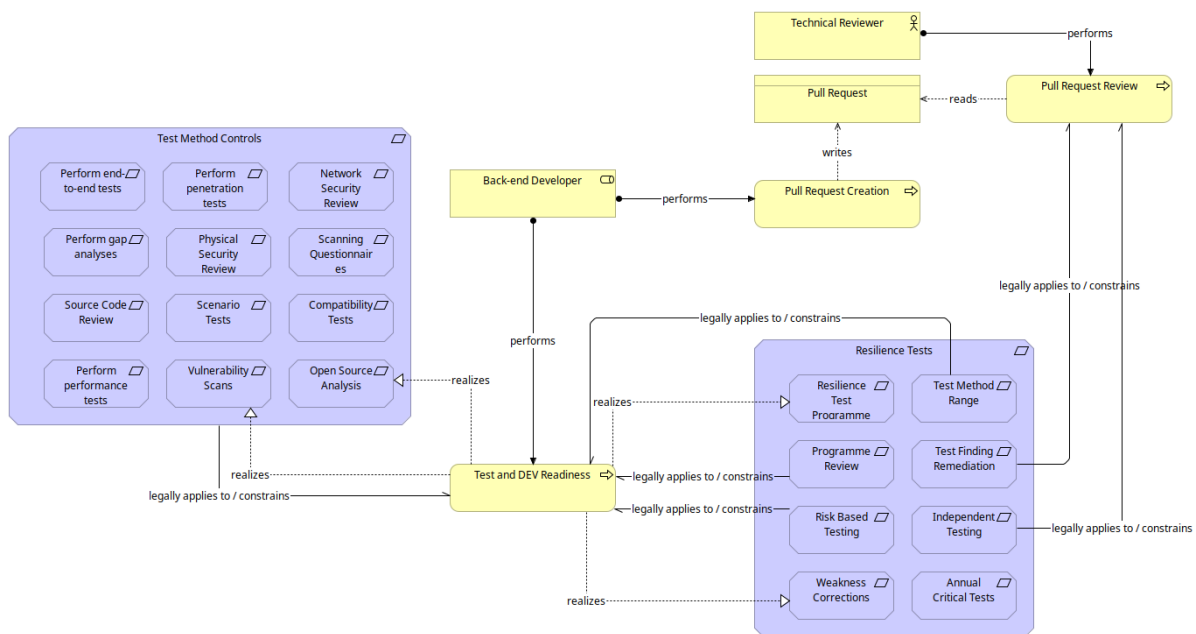


Figure 7.2: DORA Bank Developer Resilience Testing View

and did not make assumptions. The developer is not affected by these requirements, only the entity employing the developer is.

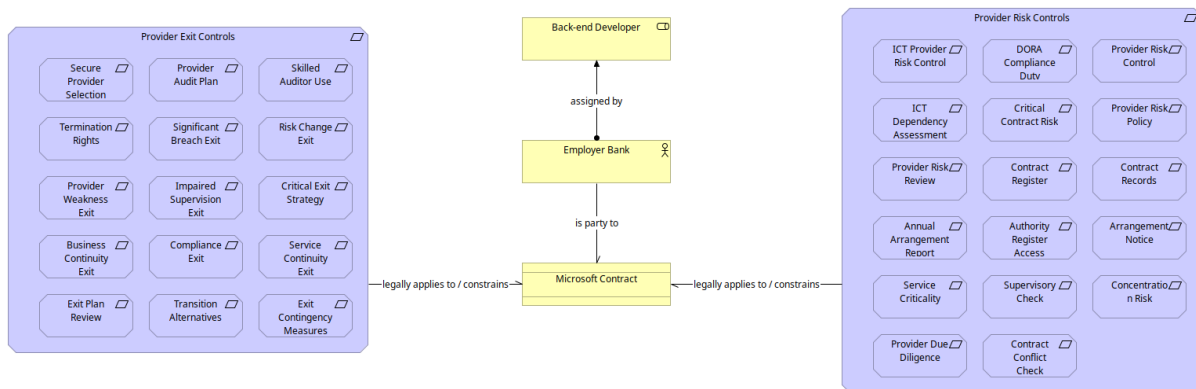


Figure 7.3: DORA Bank Developer Microsoft Cloud Risk View

A textual explanation and condensed Markdown slide diagrams complement the formal views by explaining their operational meaning and presenting their main relationships in a more accessible format. However, these initiatives have either underperformed against their objectives or demand additional development to reach their complete value. These short-comings are further addressed in 7.3.2.C

Still, the case demonstrates that the methodology can translate DORA into a stakeholder-specific AD while preserving the distinction between direct technical work, external organisational controls, and areas in which the interview evidence does not establish ownership.

7.3.2 From the Viewpoint of a Bank Infrastructure and Cloud Engineer

The second stakeholder was a Infrastructure and Cloud engineer of a banking institution, carrying database administrator responsibilities. This stakeholder had no detailed knowledge of DORA and was unfamiliar with Archimate.

Regardless, the overall feedback provided by the stakeholder was positive. In fact, the stakeholder gained a new understanding of some requirements enforced by DORA, and what processes were affected by them.

The following sub-sections review some sections of the artifacts produced by the methodology in greater detail.

7.3.2.A Operational Context and Analysis Boundary

The process began with an optimized prompt oriented towards interview building. The agent deemed it had enough information, so placeholders were not necessary. Next, an interview was generated, and sent to the stakeholder. Due to having been sent and not conducted, feedback from the stakeholder indicated the sensation that the questions were too vague. Regardless, the answers given received were rich on information and processes subject to DORA. The footprint and concerns derived from these answers were considered to be accurate by the stakeholder, and can be found in 7.3.

7.3.2.B DORA Relevance and Ownership

Regulatory relevance against DORA promptly followed the stakeholder classification. The panel outputted articles 6 through 14, 17, 18, 24 and 25. Omissions and exclusions where:

- **Article 5 — Governance and organisation:** Was omitted due to the panel never agreeing on it. This is correct, since the ICT risk control framework mentioned in it is directed to the management body, whom this stakeholder is not a part of.
- **Article 19 — Reporting of major ICT-related incidents and voluntary notification of significant cyber threats:** Excluded unanimously after debate. Also a correct decision, due to communication to competent authorities not falling within the operational footprint of the stakeholder.
- **Article 28 — General principles:** Also excluded unanimously after debate. This article establishes entity-level management of ICT third-party risk, meaning that excluding it from the relevant list of relevant requirements was also a correct decision.

The requirements outlined in the selected articles were considered accurate selections and included in the resulting relevance regulatory report.

Dimension	Case profile
Stakeholder and context	Infrastructure Engineer / Cloud Database Administrator working in the internal data technology layer of a DORA-regulated financial entity. The stakeholder administers Starburst, OpenMetadata, and SingleStore.
Core responsibilities	Provisions user access, investigates developer connection and technology-usage issues, maintains technologies through updates and configuration changes, partly automates access-control work, validates changes, and supports incident response.
Delivery path	Maintains YAML configurations through Git and VS Code; performs changes such as Starburst GitOps migration and Flux-based configuration deployment to Azure Kubernetes Service (AKS); tests updates in non-production environments; and validates changes through logs, service checks, and database-connection queries. Production changes are recorded in ServiceNow and proceed only after stability testing and the absence of reported user issues.
Key dependencies	Depends on the AKS/compute team for underlying infrastructure support, the networking team for connection issues, technology support teams for specialised correction, the security team for access-request approval, and the authorisation team for production-change approval. Key operational systems include Ranger, Flux, Azure DevOps, Azure Key Vault, Datadog, and an internal ticketing system.
Decision and incident boundary	The stakeholder's team is accountable for the technologies it configures, but the stakeholder may not alter information without an approved DevOps change. Production-change authorisation and access approval are performed by dedicated teams. During SingleStore outages, the stakeholder restarts, investigates, and mitigates the service, while escalating to the team lead, other DevOps leads, or technology support as required, and produces a post-incident report.
Evidence basis	The operational footprint contains 58 approved operational records. These comprise user-approved role context and interview-grounded tasks, technology responsibilities, dependencies, approval constraints, incident interfaces, friction points, and explicitly recorded evidence gaps.

Table 7.3: Case profile for the DORA Infrastructure and Cloud Engineer viewpoint.

After relevance resolution, pyArchimate was selected as the technology to generate the overview View. Unlike the view for the developer, instead of having separate requirement families, every requirement family was nested within a DORA Resilience Framework requirement, as can be observed in figure 7.4.

The grouping of the requirement families within the overall DORA Resilience Framework requirement is the first detail, when comparing with the previous overview view 7.1, that comes to attention.

Another difference is that while 7.1 has 134 requirements, 7.4 has only 57. However, this is a better result. The back-end developer overview view has a lot of depth, which is not intended for an overview view. These 134 requirements cover only 44 legal articles out of a 64 total. 7.4 covers 57 distinct articles, meaning each requirement is derived from an article. Since the overview is intended to be a high-level depiction of the whole regulation meant for stakeholder awareness, this is the better result amongst the two.

Both observations are attributed to the non-deterministic nature of LLMs, since nothing was done

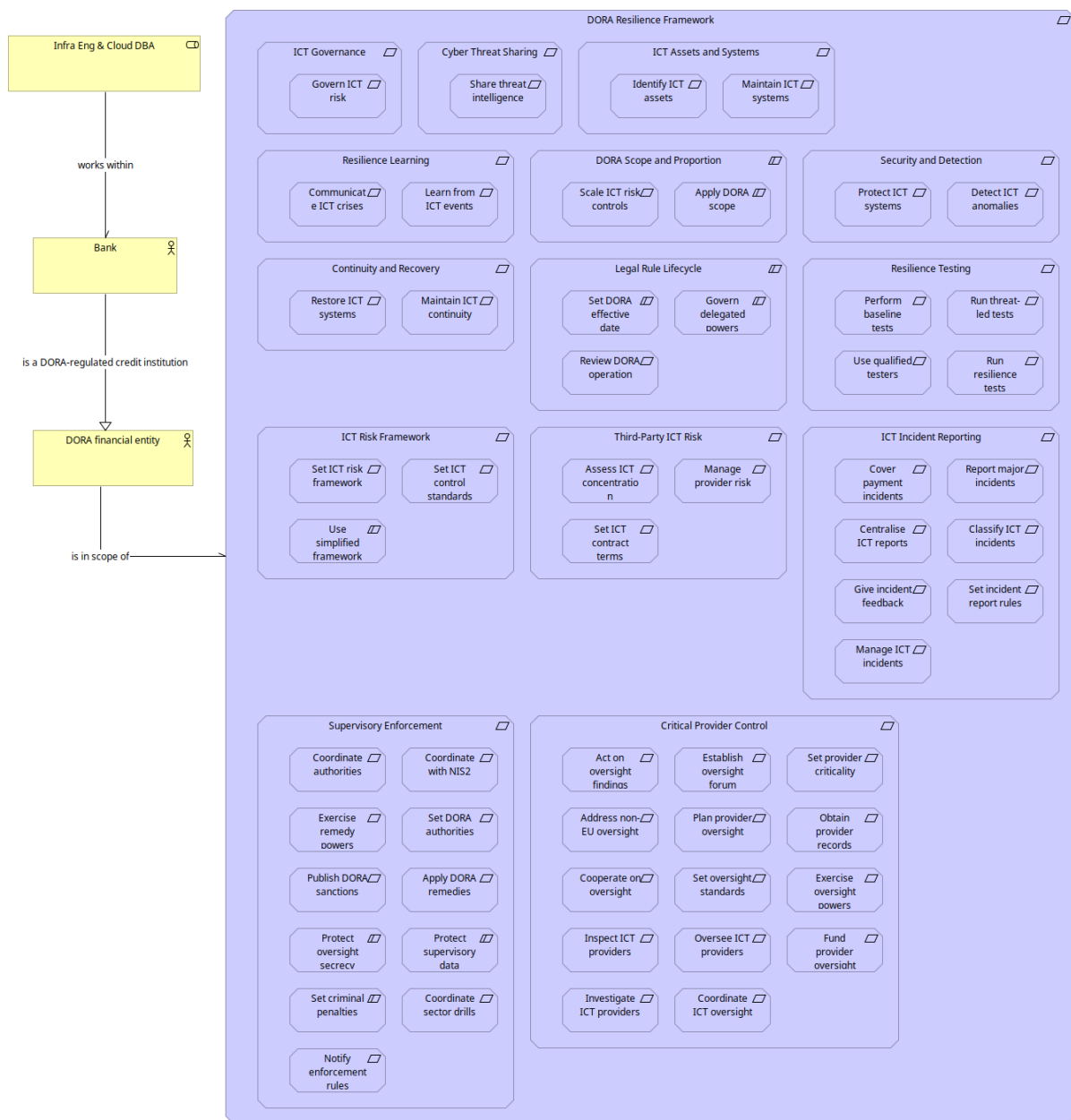


Figure 7.4: DORA Infra and Cloud Engineer Requirements Overview

differently in order to achieve distinct results.

Viewpoint scoping then organised all coverage relationships from the relevance report into fourteen legal requirement families and a role-specific operational-context stream. The scope focused on the stakeholder's administration of Starburst, OpenMetadata, and SingleStore, encompassing access control, configuration, controlled change, monitoring, incident response, recovery, and testing. Enterprise-wide governance, authority reporting, audit ownership, and other responsibilities not evidenced as belonging to the stakeholder remained outside the direct stakeholder boundary.

The resulting viewpoint, named *DORA Infrastructure Resilience*, was defined for the Infrastructure and Cloud Engineer. Its purpose is informing: it presents the relationship between the stakeholder's approved operational context and the relevant DORA requirements without assessing organisational compliance. The viewpoint addresses governance, administrative, and technical concerns through requirement families covering, among other topics, platform management, access and security, monitoring, continuity, recovery, incident handling, and resilience testing.

7.3.2.C Resulting Architecture Description

The viewpoint materialisation produced seventeen connected views: the immutable DORA overview, thirteen legal thematic views, and three operational-context views. The latter could not be connected to any specific DORA requirements, hence being modelled only as operational-context views. The thirteen legal thematic views are:

- ICT Risk Governance
- Platform & Asset Management
- Provider Dependencies
- Access & Security
- Controlled Change
- Monitoring & Detection
- Continuity Preparedness
- Recovery & Restoration
- Operational Learning
- Crisis Communication
- Incident Management

- Incident Classification
- Resilience Testing

The Controlled Change view, observable in figure 7.5 represents the approval change path for a production-change. The view was deemed accurate by the stakeholder, with possible improvements, providing preliminary stakeholder-based evidence that the second objective of the work was achieved. These observations are discussed in 7.4.

The generated textual explanation provided a per-view walk-through of every element and relation paired with an “Why it appears” explanation. The condensed Markdown slide diagrams identified presented a simplified version of every view.

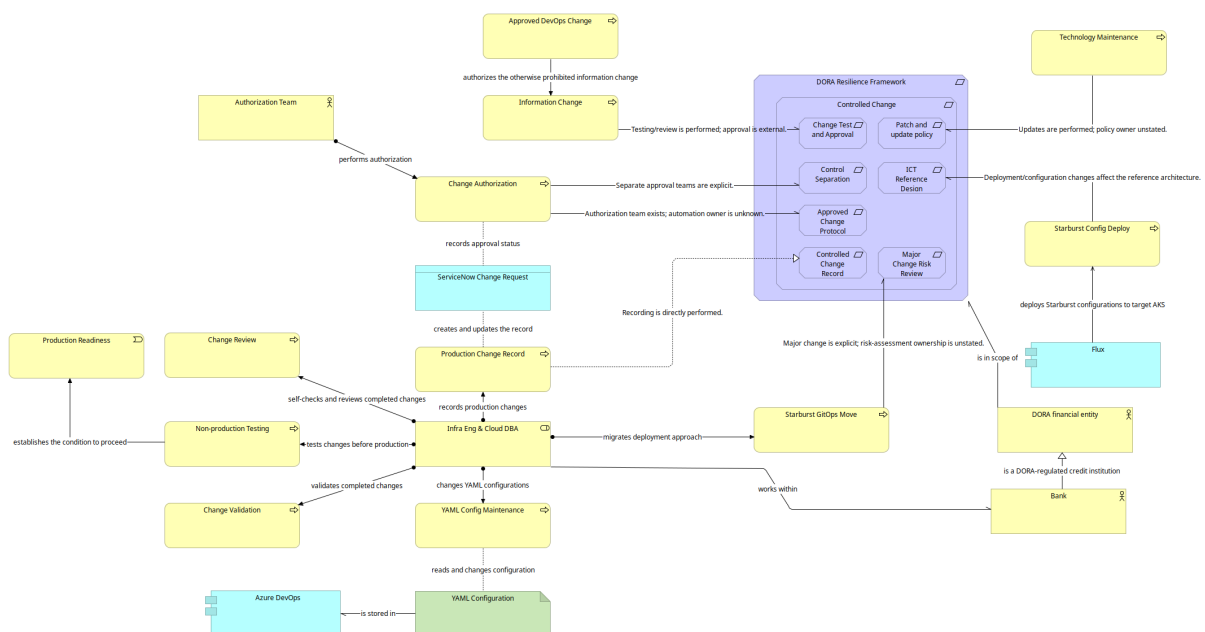


Figure 7.5: DORA Infra and Cloud Engineer "Controlled Change" View

7.4 Observations from the Demonstration

During the showing of the generated views and explanatory artifacts to their respective stakeholders, observations were made by the stakeholders regarding the ADs and registered. These form a strong basis of what is working as intended in the method, and what could be improved in the future. Regarding the views, they are as follows:

- Realization relations should be used when a requirement is realized by processes not directly performed by the stakeholder. Only using the realization relations for tasks realized directly by the stakeholder was found to be confusing.

- One major change directly realized by the stakeholder was not connected to a requirement regarding risk reviews for major changes. Upon footprint re-assessment, the risk review was left unmentioned by the stakeholder during the interview, meaning that the relation was correctly attributed as an association, and that it should be replaced with a realization.
- The stakeholder also mentioned that some elements were too abstract. What was represented in the view was correct and made sense, however what should be represented was an abstraction of the specific business process itself. (i.e., instead of “Starburst GitOps Move” should have been “Major technology move”).

Regarding the textual explanation, feedback expressed that it could be a useful idea, but the current texts need improvement. This is because the LLM currently explains *what* and *why* in the context of the methodology, and not in the operational context of the stakeholder. Examples of this can be found in every stakeholder textual explanation artifact. From the Infrastructure engineer textual explanation artifact, the provided explanation for the “Reliable ICT systems” requirement is “Atomic DORA requirement: Reliable ICT systems. It represents COV-002 under ICT Asset Controls (Art. 7)”. Although correct, the *what* does not justify what the requirement is in detail, and the *why* omits to explain what in the operational context is affected by the requirement. Improvements can be made on this subject so that the explanation provides the operational narrative needed by the stakeholder.

Feedback about the markdown slides was received poorly in the collected feedback. Although they were intended to provide an accessible summary of the formal views, their simplification frequently removed contextual detail and relationships necessary to convey the intended regulatory and operational message. As a result, they introduced a substantial loss of semantic clarity rather than improving comprehension.

Moreover, participants with no prior knowledge of ArchiMate were able to interpret the formal diagrams with relative ease when supported by the descriptions and traceability matrix. This suggests that the simplified slides do not provide sufficient additional value to justify their reduced fidelity. Future iterations of the methodology should therefore discontinue the use of these simplified slide artefacts and instead prioritise the formal ArchiMate views, complemented where necessary by stakeholder-oriented textual explanations.

8

Conclusion

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This chapter concludes the dissertation by reflecting on the development and application of the proposed method for deriving stakeholder-specific architecture viewpoints and generating their corresponding views. It summarises the main outcomes of the DORA demonstrations and discusses the considerations arising from the use of GenAI, operational evidence, and stakeholder feedback. The chapter then outlines opportunities for further development, addressing refinements to the methodology and its artefacts, extensions to the regulatory scope, and the maintenance of architectural representations as legal requirements evolve.

8.1 Outcomes and Considerations

Overall, the method presented positive outcomes in a myriad of ways. The extraction of concerns and operational footprint were faithful to the realities of the interviewed stakeholders. The selection of

relevant articles and corresponding breakdown of requirements, all the while maintaining traceability "in check" also achieved satisfactory levels.

Defined viewpoints formed a basis for view generation. The positive reception of generated views by the stakeholders is preliminary evidence of viewpoint relevance and comprehensibility. If viewpoints did not correctly define the rules for formulating stakeholder relevant views, the views themselves would not have received positive feedback.

The use of GenAI throughout the methodology highlights the importance of maintaining human judgement within the modelling process. Automated assistance supports the extraction, organisation, and representation of information, while the user reviews the proposed interpretations and approves their inclusion. Traceability and cross-artifact reconciliation support this responsibility by making the origins of represented information and the consistency of its use inspectable.

A further consideration is the dependence of stakeholder-specific viewpoints on the information made explicit during elicitation. An activity or control omitted from an interview may remain unrepresented or have unclear ownership, even when it exists in practice. The demonstration illustrated how presenting a view back to its stakeholder can reveal such omissions and prompt clarification. Generated views therefore serve both as representations of the available evidence and as instruments for enriching that evidence through discussion. The preliminary feedback supports this iterative role, while broader assessment involving additional stakeholders and independent reviewers would help establish how consistently the methodology supports understanding across different organisational and regulatory contexts.

Regardless, further improvements and refinements can be made on the current state of the methodology, as well as extending the current scope of produced artifacts. Feedback gathered from real, external stakeholders provided valuable insights regarding what produced artifacts hold the most value. The future work section (8.2) provides further insights on what could be improved on the model.

8.2 Future Work

8.2.1 Improvements on the method

In its current state, the methodology can generate structurally valid ArchiMate views, including meaningful requirement groupings and semantic containment. However, the visual layout of the generated views remains only partially automated. Although elements are positioned so that they do not overlap, their placement is not yet optimised to communicate the underlying structure clearly or to support efficient interpretation by stakeholders.

As a result, relationships may cross unrelated elements or other relationships, and long connectors may reduce the readability of the view. The generated models also do not yet make systematic use of connector bend points or routing strategies to improve the visual separation of relationships. Conse-

quently, a manual layout clean-up step is still required before the views can be considered presentation-ready.

Future work could improve either the direct model-generation mechanism or introduce a dedicated post-processing layout stage. Such work could apply layout rules based on element type, containment, relationship direction, and stakeholder reading paths, while automatically routing connectors to minimise crossings. This would reduce the manual effort required after generation, improve the readability of the resulting views, and make the methodology faster to apply in practice.

As per the received feedback indicates, the generated textual explanations of views require additional work to deliver the full scope of the potential value they hold, namely changing the current scope of explanation from the methodology to the operational footprint of the designated stakeholder.

Regarding the regulations used, this work deliberately left out regulatory and implementing technical standards, which limits the output produced by the methodology. Detailed requirements concerning, for example, ICT change management, testing environments, source-code testing, and logging are represented only where they follow directly from the text of DORA itself, and not from the supplementary technical standards. Thus, incorporation of RTS and ITS constitutes a possible extension of the methodology.

Additionally, it is worth noting that this work is in ArchiMate 3.2. Archimate 4 has been released on April of 2026, so future work must also update the agents and skills of the pipeline to produce artifacts oriented towards ArchiMate 4.

8.2.2 Traceability Matrix

The traceability matrix can still be further refined. Automation parsing currently exhibits limitations regarding amendments, and the current state of the software outputs erroneous IDs when evaluating them, due to being structured very differently from the rest of the document. This upgrade would facilitate the automated generation of a traceability matrix of a document that is mostly made-up of amendments, such as eIDAS2. Due to this, the current ID system would also need to be revisited, since in the current state it is incapable of housing enough identifiers to uniquely identify text of documents with amendments, taking into account that these introduces additional hierarchical tiers. An example of this can be seen in eIDAS2, where an original provision is immediately succeeded by its amendatory text, within the original article, something that is not representable with the current ID system. An example can be seen in figure 8.1.

Article 1
Amendments to Regulation (EU) No 910/2014
Regulation (EU) No 910/2014 is amended as follows:
(1) Article 1 is replaced by the following:

'Article 1
Subject matter

Figure 8.1: Example of an excerpt from eIDAS2 that is indescribable with the current ID system

8.2.3 Enterprise Gap Detection

While the focus of the current method is the identification and formalization of viewpoints of a stakeholder regarding an EU act, and then materializing said viewpoints into views with the purpose of informing the stakeholder about how the legislation impacts their operational footprint, the method can be taken further extended. While an interview of a single enterprise employee is not sufficient evidence to state if the company has a gap in relation to the legislation (due to the stakeholder either not being in scope of a given requirement, or simply not being informed of it), the potential to do so indicates considerable potential. In fact, a hypothesis may be that if a multitude of viewpoints from different employees indicates a gap, the gap probably exists in the enterprise. The method could be evolved into conducting interviews with multiple employees of the same role, and crossing the results in order to identify gaps within a company in regards to a certain legislation.

8.2.4 Version Aware Regulatory Maintenance

The methodology currently preserves a traceability matrix for the selected consolidated regulation. Future work could extend this capability into a version-aware maintenance mechanism. The mechanism could record the source version and effective date of the regulation, compare later consolidated versions, and identify which legal requirements, coverage records, viewpoints, and models are affected by a legal change. Rather than repeating the complete pipeline after every amendment, only the artifacts connected to changed legal provisions would require review. This would improve the long-term applicability of the methodology for regulations that evolve frequently.

Bibliography

- [1] ISO/IEC/IEEE, *ISO/IEC/IEEE 42010:2022 Software, systems and enterprise — Architecture description*, International Organization for Standardization Std., nov 2022, second edition.
- [2] Publications Office of the European Union, *Interinstitutional Style Guide*, European Union, Luxembourg, 2023, section 2: Structure of an Act. Accessed: September 19, 2026. [Online]. Available: <https://www.scribd.com/document/624373174/EU-Interinstitutional-Style-Guide>
- [3] P. A. P. Bolinhas, “Conceptual modelling of legal compliance requirements - the cases of eidas2 and nis2,” Master’s thesis, Instituto Superior Técnico, 2025.
- [4] G. Blanco-Lainé, J.-S. Sottet, and S. Dupuy-Chessa, *Using an Enterprise Architecture Model for GDPR Compliance Principles*. Springer International Publishing, 2019.
- [5] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, “A design science research methodology for information systems research,” *Journal of Management Information Systems*, vol. 24, no. 3, pp. 45–77, 2007. [Online]. Available: <https://doi.org/10.2753/MIS0742-1222240302>
- [6] A. Anaby-Tavor, D. Amid, A. Fisher, A. Bercovici, H. Ossher, M. Callery, M. Desmond, S. Krasikov, and I. Simmonds, “Insights into enterprise conceptual modeling,” *Data & Knowledge Engineering*, vol. 69, no. 12, pp. 1302–1318, 2010. [Online]. Available: <https://doi.org/10.1016/j.datak.2010.10.003>
- [7] M. M. Lankhorst, R. van Buuren, D. van Leeuwen, H. Jonkers, and H. ter Doest, “Enterprise architecture modelling—the issue of integration,” *Advanced Engineering Informatics*, vol. 18, no. 4, pp. 205–216, 2004. [Online]. Available: <https://doi.org/10.1016/j.aei.2005.01.005>
- [8] (2022) The open group standard archimate® 3.2 specification. [Online]. Available: <https://pubs.opengroup.org/architecture/archimate32-doc/singlepage/>
- [9] The Open Group. (2022) The TOGAF standard, 10th edition — introduction and core concepts. The Open Group. [Online]. Available: <https://pubs.opengroup.org/togaf-standard/introduction-and-core-concepts/chap04.html>

- [10] ISO/IEC/IEEE, *ISO/IEC/IEEE 42020:2019 Software, systems and enterprise — Architecture processes*, International Organization for Standardization Std., jul 2019, first edition.
- [11] European Commission, *European Interoperability Reference Architecture (EIRA): Chapter #2 Key Concepts and Archimate® Notation*, Interoperable Europe. [Online]. Available: <https://interoperable-europe.ec.europa.eu/collection/interoperability-architecture-solutions/solution/eira/chapter-2-key-concepts-and-archimate-notation>
- [12] S. Abualhaija, M. Ceci, N. Sannier, D. Bianculli, L. C. Briand, D. Zetsche, and M. Bodellini, “Ai-enabled regulatory change analysis of legal requirements,” in *2024 IEEE 32nd International Requirements Engineering Conference (RE)*. IEEE, 2024.
- [13] M. A. Zadenoori, J. Dąbrowski, W. Alhoshan, L. Zhao, and A. Ferrari, “Large language models (llms) for requirements engineering (re): A systematic literature review,” *arXiv preprint arXiv:2509.11446v1*, 2025.
- [14] F. Härer, “Conceptual model interpreter for large language models,” in *ER 2023: Companion Proceedings of the 42nd International Conference on Conceptual Modeling*. CEUR Workshop Proceedings, 2023.
- [15] C. Jeong, “A study on the implementation of generative ai services using an enterprise data-based llm application architecture,” *arXiv preprint arXiv:2309.01105v2*, 2023. [Online]. Available: <https://arxiv.org/abs/2309.01105>
- [16] European Union, “Consolidated version of the treaty on european union – article 5: Principle of conferral,” Official Journal of the European Union, C 202, p. 18, Jun. 2016, cELEX: 12016M005. [Online]. Available: <https://eur-lex.europa.eu/legal-content/EN/TXT/HTML/?uri=CELEX:12016M005>
- [17] T. D. Breux and A. I. Antón, “Analyzing regulatory rules for privacy and security requirements,” *IEEE Transactions on Software Engineering*, Jan. 2008. [Online]. Available: <https://doi.org/10.1109/TSE.2007.70746>
- [18] (2022) Regulation (eu) 2022/2554 of the european parliament and of the council of 14 december 2022 on digital operational resilience for the financial sector and amending regulations (ec) no 1060/2009, (eu) no 648/2012, (eu) no 600/2014, (eu) no 909/2014 and (eu) 2016/1011. European Union. [Online]. Available: <https://eur-lex.europa.eu/eli/reg/2022/2554/oj/eng>
- [19] Publications Office of the European Union. 2. structure of a legal act. [Online]. Available: <https://style-guide.europa.eu/en/content/-/isg/topic?identifier=2-structure-legal-act>

- [20] G. M. P. Karakasilioti, "Supporting the digital operational resilience of the financial sector: The eu's dora digital operational resilience act," Master's thesis, UNIVERSITY OF PIRAEUS, Piraeus, Greece, 2024.
- [21] M. A. Umar and K. Lano, "Advances in automated support for requirements engineering: a systematic literature review," *Requirements Engineering*, vol. 29, pp. 177–207, 2024.
- [22] P. Elahidoost, "Towards a tool supported approach for regulatory requirements engineering," in *2024 IEEE 32nd International Requirements Engineering Conference (RE)*. Reykjavik, Iceland: IEEE, Jun. 2024. [Online]. Available: <https://doi.org/10.1109/RE59067.2024.00067>
- [23] Y. Zhou, S. Wang, Y. Su, W. Du, Y. Fang, and X. Lin, "A comprehensive survey on agent skills: Taxonomy, techniques, and applications," *arXiv preprint arXiv:2605.07358*, 2026. [Online]. Available: <https://arxiv.org/abs/2605.07358>
- [24] R. Xu and Y. Yan, "Agent skills for large language models: Architecture, acquisition, security, and the path forward," in *Proceedings of the ACM CAIS 2026 Workshop on AgentSkills*, 2026, pp. 1–9. [Online]. Available: <https://arxiv.org/abs/2602.12430>
- [25] M. Oriol, Q. Motger, J. Marco, and X. Franch, "Multi-agent debate strategies to enhance requirements engineering with large language models," Jul. 2025. [Online]. Available: <https://arxiv.org/abs/2507.05981>
- [26] S. Gómez, A. Mozo, T. Navarro, S. Gálvez, and F. L. Valverde, "Multi-agent debate system based on large language models: Structured deliberation and validation in satellite communications," Research Square preprint, May 2026, posted May 11, 2026. [Online]. Available: <https://doi.org/10.21203/rs.3.rs-9575030/v1>
- [27] O. Kosenkov, M. Unterkalmsteiner, D. Mendez, and J. Fischbach, "Regulatory requirements engineering in large enterprises: An interview study on the european accessibility act," in *Product-Focused Software Process Improvement: 25th International Conference, PROFES 2024*, ser. Lecture Notes in Computer Science, D. Pfahl, J. Gonzalez Huerta, J. Klünder, and H. Anwar, Eds., vol. 15452. Springer, 2025, pp. 204–220. [Online]. Available: https://doi.org/10.1007/978-3-031-78386-9_14
- [28] T. D. Breaux, A. I. Antón, K. Boucher, and M. Dorfman, "Legal requirements, compliance and practice: An industry case study in accessibility," in *Proceedings of the 16th IEEE International Requirements Engineering Conference (RE 2008)*. IEEE, 2008, pp. 43–52. [Online]. Available: <https://doi.org/10.1109/RE.2008.36>

- [29] S. Ghanavati, A. Rifaut, and E. Dubois, "Goal-oriented compliance with multiple regulations," in *2014 IEEE 22nd International Requirements Engineering Conference (RE)*. IEEE, 2014.
- [30] C. Autio, R. Schwartz, J. Dunietz, S. Jain, M. Stanley, E. Tabassi, P. Hall, and K. Roberts, "Artificial intelligence risk management framework: Generative artificial intelligence profile," National Institute of Standards and Technology, Tech. Rep. NIST AI 600-1, 2024. [Online]. Available: <https://doi.org/10.6028/NIST.AI.600-1>



Pipeline Overview

Bellow is a table that summarizes the pipeline, along with every agent and skill involved in each step, as well as the outputted artifact. For further detail, please refer to the following GitHub: https://github.com/AntonioBrasIST/Architecture_Viewpoints_for_EU_Regulation

Step	Step name	Agents	Skills	Artifact
1	Prompt Optimization	0-prompt-maker	prompt-maker	1_optimized_prompt.md
2	Produce Interview	1-interview-designer	regulation-blind-operational-discovery	2_interview.md
–	Manual Elicitation	User	Manual elicitation and response recording	2_interview.md
3	Context Discovery	2-operational-footprint-extractor	extract-operational-footprint; evidence-grounded-relationship-ledger	3_operational_footprint.md
4	Stakeholder & Concern ID	3-stakeholder-classifier	extract-stakeholder-class-and-concerns	4_stakeholder_classification.md
5	Regulatory Relevance Analysis	4-eu-legislation-identifier; 4a, 4b, 4c, 4d	legislation-relevance-evaluator; regulatory-consensus-moderator; legal-text-anchor-resolver; stakeholder-footprint-coverage; evidence-grounded-relationship-ledger; stakeholder-legal-role-mapper	5_regulatory_relevance.md; <REGULATION>-CELEX:<CELEX>.xlsx
–	Technology Selection Gate	–	–	–
5a	Contextualized Regulation Overview	Regu- 4e-a, 4e-b, or 4e-c overview modeller	regulation-overview-designer; legal-text-anchor-resolver; stakeholder-legal-role-mapper; regulatory-view-structure; connected-view-validator; selected native-model skills	Views/0_<law-slug>_regulation_overview.*
6	Viewpoint Scope Justification	& 5-stakeholder-viewpoint-scooper	viewpoint-scope-definer; stakeholder-footprint-coverage; evidence-grounded-relationship-ledger; regulatory-view-structure; connected-view-validator	6_viewpoint_scope.md
7	Viewpoint Generation	6-viewpoint-materializer	viewpoint-creator; viewpoint-materializer; stakeholder-footprint-coverage; evidence-grounded-relationship-ledger; regulatory-view-structure; connected-view-validator	7_viewpoint.md; 7_view_graph.json
8	Textual View Explanation	7-view-narrative-explainer	view-narrative-explainer; stakeholder-footprint-coverage; evidence-grounded-relationship-ledger; regulatory-view-structure; connected-view-validator	Views/1_view_explanation.md
9	Markdown Slides	Dia- 8-informal-diagram-designer	informal-markdown-diagram-designer; stakeholder-footprint-coverage; evidence-grounded-relationship-ledger; regulatory-view-structure; connected-view-validator	Views/2_slide_diagrams.md
10a	ArchiMate XML Generation	9a-archimate-model-generator	archimate-xml-generator; archimate-symbol-validator; stakeholder-footprint-coverage; evidence-grounded-relationship-ledger; regulatory-view-structure; connected-view-validator	Views/3_view_model.archimate
10b	LikeC4 Model Generation	9b-likec4-model-generator	likec4-dsl; stakeholder-footprint-coverage; evidence-grounded-relationship-ledger; regulatory-view-structure; connected-view-validator	Views/3_view_model.c4
10c	pyArchiMate Generation	9c-pyarchimate-model-generator	pyarchimate-syntax-reference; archimate-symbol-validator; stakeholder-footprint-coverage; evidence-grounded-relationship-ledger; regulatory-view-structure; connected-view-validator; pyarchimate-model-generator	3_view_model_pyarchimate.py; Views/3_view_model_pyarchimate.archimate
11	Cross-Artifact Reconciliation	Recon- 10-view-cross-artifact-reconciler	view-cross-artifact-reconciler; legal-text-anchor-resolver; stakeholder-footprint-coverage; evidence-grounded-relationship-ledger; regulatory-view-structure; connected-view-validator	Audit report; synchronized view updates

